Half-degree irreducibles over $\mathbb{F}_2$: barriers, negative results, and what to try next

Michael Bommarito

23 August 2026

Status. *Nothing in this document proves the Kaser–Lemire conjecture.* Several of the results recorded here are textbook theorems applied to new data, and they are labelled as such. The purpose is to say precisely what has been tried, what is provably blocked, and what a specialist might still attack.

Abstract

For every n, is there a monic irreducible $f \in \mathbb{F}_2[x]$ of degree n with $\deg(f - x^n) \le \lfloor n/2 \rfloor$? The question is Lemire's (MathOverflow, 2011), and Kaser and Lemire published it as a conjecture in 2014 with a Barrett-reduction motivation. Reversing f turns it into Legendre's conjecture for $\mathbb{F}_2[t]$: a prime in the interval of length $2\sqrt{X}$ around $X = 2^n$. The Riemann hypothesis is a theorem in this setting (Weil), and it is short of the conjecture by exactly one logarithm. This document is the record of a systematic attack on that logarithm. We collect what is unconditionally proved (an almost-all theorem with a sharp constant; an infinite family for every in-window seed; the exact sieve output of the window; certified witnesses to $n = 3000$), and then six barriers, each stated as a proposition with proof and exact scope, that between them eliminate moduli-only Fourier arguments, group actions, the known irreducibility-preserving construction toolbox, all lower-bound sieves at the window's own level of distribution, Betti-number bounds on Katz's parameter space, and the one available fixed-q *disproof* template. We then re-pose the surviving geometric question — a degree/weight statement for the Adams twist of Katz's universal sheaf on Prim_j — and report exact L-function computations that make it look alive rather than dead, together with the two named open statements it reduces to. A short marked section records the working method, including five occasions on which our own claims were wrong.

Contents

1 Introduction

1.1 The conjecture and where it comes from

Write $W_n = \{x^n + g \ : \ g \in \mathbb{F}_2[x],\ \deg g \le \lfloor n/2 \rfloor\}$ for the *half-degree window* at degree n; $|W_n| = 2^{\lfloor n/2 \rfloor + 1}$.

Conjecture (Kaser–Lemire). For every $n \ge 1$ the window W_n contains a monic irreducible polynomial; equivalently, there is a monic irreducible $f \in \mathbb{F}_2[x]$ of degree n with $\deg(f - x^n) \le \lfloor n/2 \rfloor$.

The question was posed publicly by Daniel Lemire on MathOverflow on 23 November 2011, under the title "Can we always find such an irreducible polynomial of degree n where degree$(p(x) - x^n) \le n/2$?" [1]. It appears as a conjecture in print in Kaser and Lemire, *Strongly universal string hashing is fast* [2], in the section "GF Multilinear": they verify it for $L \in \{1, \ldots, 400\}$ against Arndt's table [3] and conjecture it in general. The motivation is computational. In a string-hashing kernel one reduces a product modulo an irreducible f of degree n over $\mathbb{F}_2$; Barrett reduction is cheapest when the non-leading part of f is confined to the bottom half of the coefficient range, because then a single carry-less multiplication by the precomputed reciprocal suffices and the correction step does not overflow. So the conjecture is exactly the assertion that this optimization is always available.

The 2011 thread already contains most of the mathematical landscape. Elkies and Zaimi both state the expected truth — that the minimal *subdegree* $\deg(f - x^n)$ should be $O(\log n)$, not $n/2$ — and Arndt's table bears this out: the minimal subdegree is at most 10 for every $n \le 400$. Ellenberg gives the framing that organizes everything below: this is Legendre's conjecture for $\mathbb{F}_2[t]$, and Cramér's heuristic under RH buys intervals of length about $\sqrt{X}\log X$, i.e. subdegree $n/2 + \log n$, one logarithm on the wrong side. Voloch notes that Cohen's prescribed-coefficient criterion likewise stops at $m < n/2 + O(\log n)$. And Emil Jeřábek's comment gives the infinite family $x^{2\cdot 3^k} + x^{3^k} + 1 = \Phi_{3^{k+1}}$, which is an exercise in Lidl–Niederreiter, so that the degrees $n = 2 \cdot 3^k$ have been settled since 2011.

1.2 The one-line reason it is hard

Reversing $f \mapsto f^*(T) = T^{\deg f} f(1/T)$ converts the window condition into a congruence (Section 2): W_n contains an irreducible if and only if there is a prime $P \in \mathbb{F}_2[T]$ of degree n with $P \equiv 1 \pmod{T^{\lceil n/2 \rceil}}$. Thus the conjecture is *Legendre's conjecture for* $\mathbb{F}_2[t]$: a prime in a short interval of length $2^{\lceil n/2 \rceil}$, i.e. between $\sqrt{X}$ and $2\sqrt{X}$ where $X = 2^n$. Equivalently it is Linnik's problem with exponent exactly 2, for the residue 1 and the prime-power modulus $T^{\lceil n/2 \rceil}$.

Over $\mathbb{Z}$ this is hopeless because RH is unavailable. Here RH *is* available: it is Weil's theorem, and it gives, for every Hayes character χ of conductor j,

$$|S_n(\chi)| \ :=\ \Big| \sum_{\deg F = n} \Lambda(F)\chi(\langle F \rangle_j) \Big| \ \le\ (j-1)\, 2^{n/2}.$$

Summing this over the 2^ℓ characters, with $\ell = \lceil n/2 \rceil - 1$, gives a main term $2^{n-\ell}$ against an error $\sum_j 2^{j-1}(j-1)2^{n/2} \approx \ell 2^\ell 2^{n/2}$, and the two are the same size up to the factor ℓ. That factor is the whole problem. The classical statement of the same phenomenon: the top $n/2 - \log_2 n$ coefficients of an irreducible are prescribable unconditionally (Hayes plus Weil, in the sharp form of Hsu [29] = Cohen [30]), and Kaser–Lemire asks for $\lceil n/2\rceil - 1$ of them. **The conjecture is exactly the removal of the** log **that RH leaves behind.** It is not a power gap; it is the classical square-root barrier, transplanted to a setting where the square-root barrier is a theorem rather than a hypothesis.

1.3 What this document is, and is not

It is a companion to the three-page roadmap note [33], which states the reduction chain and the single open estimate, and to the almost-all note [34]. Where those are compressed, this one is expansive: it records the negative results, which are the part of the work that is most likely to save a reader time.

Three things it is not. It is not a proof, and it does not contain a plausible sketch of one. It is not a survey of the function-field literature: papers are cited only where they bear on this window at $q = 2$, and the fixed-q, growing-conductor regime is essentially untouched in print. And it is not a claim of novelty for most of what it proves: the almost-all theorem is Parseval+Weil+Chebyshev, the infinite family is Lidl–Niederreiter [7] Theorem 3.35 applied to a certified seed ledger, and the sieve output is Jurkat–Richert applied to a sequence with exact divisor data. Each such statement is labelled where it appears.

Status tags. Every substantive claim below carries one of five tags, and the distinctions are enforced rather than decorative:

- **PROVED** — a theorem, with a proof here or an exact citation.
- **FINITE** — machine-checked finite evidence: an exact computation over a stated finite range, re-verified by an independent engine and gated by a checker that exits nonzero on failure.
- **MEASURED** — a measurement or a heuristic model fit; a number, not a theorem, and not a proof of the asymptotic it suggests.
- **OPEN** — open, with the precise statement given.
- **CONJECTURED** — conjectured here or in the literature, with the source.

Internal sources are cited by lane note number ("note 13"); these are the research notes in `docs/research/10-cas/lemire-signed-trace/` of the repository [35], together with the checkers in `scripts/lemire-signed-trace/`. Where a claim is machine-checked, the script that checks it is named.

2 Setting and the reduction

This section is compressed; the full chain, with the endpoint bookkeeping, is in the roadmap note [33]. Everything in it is **PROVED**.

The ray class group. For $j \geq 0$ put

$$\mathcal{E}_j \;=\; (1 + x\mathbb{F}_2[x])/(x^{j+1}) \;=\; \left(\mathbb{F}_2[x]/x^{j+1}\right)^\times, \qquad |\mathcal{E}_j| = 2^j,$$

and for monic F let $\langle F\rangle_j = x^{\deg F} F(x^{-1}) \bmod x^{j+1}$. With $\ell = \lceil n/2\rceil - 1$, the window condition $\deg(f - x^n) \leq \lfloor n/2\rfloor$ is exactly $\langle f\rangle_\ell = 1$. So Kaser–Lemire asks for a prime of degree n in the *identity ray class* of $\mathcal{E}_\ell$.

By Artin–Hasse, $\mathcal{E}_j$ is the truncated big Witt ring of $\mathbb{F}_2$ and splits (Katz [11], Lemma 2.2) as $\mathcal{E}_j \cong \prod_{k \text{ odd} \leq j} \mathbb{Z}/2^{e_k}$ with $e_k = \lfloor \log_2(j/k)\rfloor + 1$ and $\sum_k e_k = j$. Under this splitting the class of $\alpha \in \mathbb{F}_{2^n}$ is the vector of Galois-ring traces $\mathrm{Tr}_{\mathrm{GR}(2^{e_k},n)}(\tilde{\alpha}^k) \bmod 2^{e_k}$ of the Teichmüller lift, so the identity class is the locus where *all odd-power traces vanish to prescribed dyadic precisions.* This is the description that makes the class genuinely distinguished, and every barrier in Section 4 is in the end a statement about that.

Hayes characters and the Weil bound. The characters of $\mathcal{E}_j$ are the even Hayes characters modulo x^{j+1} [4]. For χ of exact conductor j, $L(\chi, T) = \prod_{i \leq j-1}(1 - \alpha_i T)$ is a polynomial of degree $j - 1$, pure of weight one ($|\alpha_i| = \sqrt{2}$) by Weil [5], and

$$S_n(\chi) = \sum_{\deg F = n} \Lambda(F)\chi(\langle F\rangle_j) = -\sum_i \alpha_i^n, \qquad |S_n(\chi)| \leq (j-1)2^{n/2}.$$

Conductor 1 gives $L = 1$, hence $S_n = 0$. Exactly 2^{j-1} characters have exact conductor j.

The telescope and (REL). Write $N_j(g) = \sum_{\deg F=n,\ \langle F\rangle_j = g} \Lambda(F)$. Binary splitting of the two-element fibres of $\mathcal{E}_j \to \mathcal{E}_{j-1}$ gives the Haar telescope $2^\ell N_\ell(e) = 2^n + \sum_{j\le \ell} \varepsilon_j(e) 2^{j-1} H_j(b_j(e))$ with H_j the difference of the two child populations. Put $c = \lceil \log_2 \ell \rceil$, $a = \ell - c - 1$,

$$C_{\ell,n} = \sum_{j=a}^{\ell} 2^{j-1} H_j(1), \qquad B_{\ell,n} = 2^{2\ell} - 2^{\lceil n/2\rceil} \sum_{1\le j<a} (j-1)2^{j-1} \ > \ 0.$$

Summing the large conductors *before* taking absolute values telescopes them, and the proper-power count at both endpoints reduces the conjecture to

$$\text{(REL)} \qquad C_{\ell,n} \ > \ -B_{\ell,n}.$$

Order layers and (HWO). Let $2^s\mathcal{E}_j$ be the 2^s-power subgroup, $P_{j,s} = \sum_{g\in 2^s\mathcal{E}_j} N_j(g)$, $h_{j,s} = 2^{\,j-\lfloor j/2^s\rfloor}$, and

$$T_{j,s}(n) \ = \ h_{j,s}P_{j,s} - h_{j,s-1}P_{j,s-1} - h_{j-1,s}P_{j-1,s} + h_{j-1,s-1}P_{j-1,s-1}$$

the sum of $S_n(\chi)$ over the $\#X_{j,s}$ characters of exact conductor j and exact order 2^s. With Q the largest power of two with $3Q\lceil \log_2 \ell\rceil \le \ell$, the Weil bound pays for every layer of order at most Q, and what remains is

$$\text{(HWO)} \qquad 4\ell\,|T_{j,s}(n)| \ \le \ \#X_{j,s}\,(j-1)\,2^{\lceil n/2\rceil} \qquad (a \le j \le \ell,\ 2^s > Q),$$

i.e. a factor 4ℓ of saving over Weil for the *signed* sum over a complete layer, uniformly in ℓ. **(HWO) $\Rightarrow$ (REL) $\Rightarrow$ Kaser–Lemire, and this implication is proved** (note 01; endpoint ledger replayed for $200 \le \ell \le 1024$). A necessary conductor/order case split makes (HWO) equivalent to a sparse-discrepancy statement: with $d_s = \lfloor (j-1)/2^s \rfloor$, $\Delta_{j,s} = 2P_{j,s} - P_{j-1,s}$ and $R = 2^{d_{s-1}-d_s}$,

$$\text{(NSD)}\ \ 4\ell|R\Delta_{j,s} - \Delta_{j,s-1}| \le (R-1)(j-1)2^{\lceil n/2\rceil} \quad \text{when } 2^{s-1} \nmid j,$$

$$\text{(RSD)}\ \ 4\ell|\Delta_{j,s}| \le (j-1)2^{\lceil n/2\rceil} \quad \text{when } 2^{s-1} \mid j,\ 2^s \nmid j.$$

The resonant case is not a small-scale artefact: at $j = \ell = 200$ the order $2^s = 16$ is high and $8 \mid 200$, $16 \nmid 200$.

The cylinder form (CYL). Let $K = \ker(\mathcal{E}_\ell \to \mathcal{E}_{a-1})$, of order $2^{\ell-a+1} = 2^{c+2}$, so that

$$4\ell \ \le \ |K| \ < \ 8\ell. \tag{1}$$

For $\psi \in \widehat{K}$ put $A_\psi = \sum_{g\in K} N(g)\psi(g)$, the prime mass of the cylinder $\{x^n + g : \deg g \le n-a\}$ twisted by a sign pattern of the $c+2$ coefficients of $x^{n-a}, \dots, x^{n-\ell}$. Fourier inversion on K gives

$$\text{(CYL)} \qquad |A_\psi| < 2^{\ell-1} \ \text{ for every } \psi \ne 1 \qquad \Longrightarrow \qquad \text{(REL)}.$$

(CYL) is the weakest sufficient statement currently known, and it is the ledger's open fact `F:gf2-lemire-cylinder-twist-sup-bound`. Weil gives only $|A_\psi| \le (\ell-1)2^{\lceil n/2\rceil}$, a factor $4(\ell-1)$ short.

The reversal duality, with the index. Keating–Rudnick's involution [26] ($f \mapsto f^*$, $\Lambda(f^*) = \Lambda(f)$ when $f(0) \ne 0$) gives the bijection

$$\{f \text{ monic irred.},\ \deg f = n,\ \deg(f - T^n) \le \lfloor n/2\rfloor\} \ \longleftrightarrow \ \{P \text{ monic irred.},\ \deg P = n,\ P \equiv 1 \pmod{T^r}\},$$

with

$$\boxed{r \ = \ n - \lfloor n/2\rfloor \ = \ \lceil n/2\rceil \ = \ \ell + 1}$$

PROVED (note 16; `lemire_largeq.py`, verified on 89 pairs (q,n) and 24,090 polynomials, $q \in \{2,3,4,5,7,8,9\}$, $n \le 24$).

Remark 2.1 (the off-by-one is not cosmetic)**.** The naive index $\lfloor n/2\rfloor + 1$ agrees with $\lceil n/2\rceil$ at odd n and is *one too large* at even n. This changes the level of distribution required from exactly $1/2$ to $1/2 + 1/n$, hence whether $\omega = 1/2$ is admissible at all; and it can change the answer. Over $\mathbb{F}_3$ at $n = 4$ there are 6 in-window irreducibles, 2 of them with $f(0) = 1$, matching the progression count at $r = 2$; at $r = 3$ the progression is *empty*. An argument run at the wrong index would have concluded that no in-window irreducible of degree 4 exists over $\mathbb{F}_3$, which is false. (Note 16; the checker's mutation control changes r and kills exactly the three duality checks.)

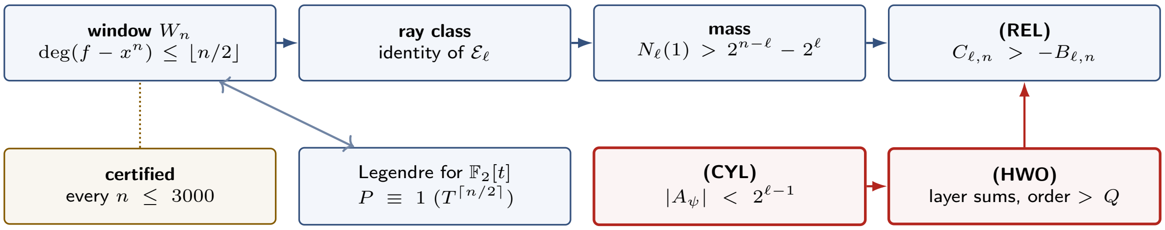

Figure 1: The reduction chain. Blue arrows are proved implications; the two red boxes are the open sufficient statements, either of which closes the chain. The gold box is the finite verification, which is insurance and not evidence for the open step. The double arrow records that the window question *is* Legendre's conjecture for $\mathbb{F}_2[t]$, so no argument blind to the centre of the interval can separate them (Section 4, Barrier IV).

3 What is proved

3.1 Certified witnesses to $n = 3000$

Proposition 3.1 (FINITE). *For every $n \le 3000$ the window W_n contains an explicit irreducible polynomial, doubly certified.*

For $401 \le n \le 3000$ each witness is sparse, $x^n + \sum x^{e_i} + 1$ with every $e_i \le \lfloor n/2 \rfloor$, and carries a Frobenius/Bézout irreducibility certificate that is re-verified by an independent checker: 2600 of 2600 pass, 1334 trinomials and 1266 pentanomials. The replayable table (degree, tail exponents, SHA-256 of the artifact, check status) is `data/witnesses-401-3000-sha256.tsv`. A third, independent flint-based search with pure-Python Rabin re-verification agrees on existence for every degree in 401..842, producing identical trinomial witnesses (note 02 §6; `lemire_witness_search.py`).

Two honest remarks. First, powers of two need denser witnesses: Swan's theorem forbids an irreducible trinomial of any degree divisible by 8, which is why roughly half the table is pentanomials. Second, and more important, this is *insurance, not evidence*: combined with the reduction, the chain is complete for $n \le 3000$ and for all n once (HWO) or (CYL) holds for $\ell \ge 1500$. It says nothing about the open estimate.

3.2 The almost-all theorem

Theorem 3.2 (PROVED; note 05, `lemire_almostall.py`). *Let $\ell \ge 1$, $n \in \{2\ell+1, 2\ell+2\}$, $0 < t \le 1$, and put $\varepsilon(\ell) = (\ell^2 - 4\ell + 6) - 6 \cdot 2^{-\ell}$ and $\kappa_n = 2^{2\lceil n/2 \rceil - n} \in \{1, 2\}$. Then*

$$\#\{g \in \mathcal{E}_\ell \ :\ |N_\ell(g) - 2^{n-\ell}| \ge t\, 2^{n-\ell}\} \ \le\ \kappa_n\, 2^{2\ell-n}\, \varepsilon(\ell)/t^2.$$

Consequently, for $\ell \ge 5$, the number of top-half patterns $h \in \mathbb{F}_2^\ell$ carried by no *irreducible of degree n is at most $4(\ell^2 - 4\ell + 6) < 4\ell^2$ for odd n and at most $\ell^2 - 4\ell + 6 < \ell^2$ for even n; so all but a fraction $< 4\ell^2 2^{-\ell}$ of the 2^ℓ patterns are realized.*

Proof sketch. Parseval on $\mathcal{E}_\ell$ turns $\sum_g (N_\ell(g) - 2^{n-\ell})^2$ into $2^{-\ell} \sum_{\chi \ne 1} |S_n(\chi)|^2$; Weil and the conductor count give $\sum_{\chi \ne 1} |S_n|^2 \le 2^{2\lceil n/2 \rceil} \sum_{j=1}^{\ell} 2^{j-1}(j-1)^2 = 2^{2\lceil n/2 \rceil} A(\ell)$ with $A(\ell) = \sum_{i=1}^{\ell-1} i^2 2^i = 2^\ell(\ell^2 - 4\ell + 6) - 6$ (induction); Chebyshev finishes. The irreducible count follows after a Markov bound on the proper-power mass, which is empty for odd n and has at most one exceptional class for even n. □

Honest scope. This is Parseval + Weil + Chebyshev: the function-field Barban–Davenport–Halberstam second moment, unconditional here only because RH is Weil's theorem. The method is standard; the endpoint statement with explicit constants at fixed $q = 2$ we did not find in print, and it should be treated as folklore made explicit, not as new. Hayes [4] and Hsu [29] give *pointwise* asymptotics for every class while $\ell \le n/2 - \log_2 \ell - O(1)$ — stronger in that range, failing exactly at the last $\log_2 \ell$ coefficients, which is the entire gap.

What it measures. The exceptional set is *empty* at every size computed ($\ell \le 24$), while the theorem allows 10% of classes at $\ell = 12$; the worst class is 0.3–13% off the mean, and $\min_g N_\ell(g)/\text{mean} = 0.9971$ at $(\ell, n) = (24, 50)$ (FINITE). The observed second moment is at its Sato–Tate value to three digits from $\ell = 16$ on, so Weil is lossy here by exactly the expected factor $\sim \ell$ (MEASURED). Per conductor, the second moment *saturates* the sharp Weil bound

seed f	m	e	$A(f)$	degrees reached
x^2+x+1	2	3	$\{3\}$	$n = 2 \cdot 3^k$
x^3+x+1	3	7	$\{7\}$	$n = 3 \cdot 7^k$ (**odd**)
x^4+x+1	4	15	$\{3,5\}$	$n = 4 \cdot 3^a 5^b$
x^5+x^2+1	5	31	$\{31\}$	$n = 5 \cdot 31^k$ (**odd**)
x^6+x+1	6	63	$\{3,7\}$	$n = 6 \cdot 3^a 7^b$
$x^8+x^4+x^3+x^2+1$	8	255	$\{3,5,17\}$	$n = 8 \cdot 3^a 5^b 17^c$
x^9+x^4+1	9	511	$\{7,73\}$	$n = 9 \cdot 7^a 73^b$ (**odd**)
$x^{12}+x^6+x^4+x+1$	12	4095	$\{3,5,7,13\}$	$n = 12 \cdot 3^a 5^b 7^c 13^d$

Table 1: Seeds and the degrees they reach (note 08). Explicit odd witnesses, verified irreducible and in-window by two independent engines: $x^{21}+x^7+1$, $x^{147}+x^{49}+1$, $x^{1029}+x^{343}+1$, $x^{7203}+x^{2401}+1$ ($m=3$), and $x^{155}+x^{62}+1$ ($m=5$, $t=31$).

at $j=2$ (ratio 1.00000 at every endpoint) and sits at $\mathrm{rms}|S|/((j-1)2^{n/2}) = 1/\sqrt{j-1}$ to four digits at $j=24$: Weil is attained at the stabilizer boundary and $\sqrt{j-1}$-lossy above it.

The residual. Kaser–Lemire is exactly the statement that *one named pattern*, $h=0$, is not exceptional. Theorem 3.2 cannot address it, and not for want of constants: its three inputs are invariant under permutations of $\mathcal{E}_\ell$. Barrier I (§4.1) makes that precise.

3.3 The monomial-composition family

Theorem 3.3 (**PROVED**; note 08 Theorem A, `lemire_composition_family.py`). *Let $f \in \mathbb{F}_2[x]$ be irreducible of degree m with $f(0) \neq 0$, of order $e = \mathrm{ord}(f)$, and suppose f is in-window, i.e.* $\deg(f - x^m) \leq \lfloor m/2 \rfloor$. *Let $t \geq 2$ satisfy*

(i) $\mathrm{rad}(t) \mid e$*: every prime divisor of t divides e; and*

(ii) $\gcd(t, (2^m-1)/e) = 1$.

Then $f(x^t)$ is an in-window irreducible of degree mt and order et. Hence Kaser–Lemire holds at $n = mt$.

Proof. Irreducibility is Lidl–Niederreiter [7] Theorem 3.35 (stated there in exhaustive form; the single-polynomial "iff" form is Handbook of Finite Fields [8] Thm. 3.2.5, after Menezes et al.). At $q=2$ the fourth hypothesis of the book's statement ($q^m \equiv 1 \bmod 4$ when $4 \mid t$) is vacuous, because $e \mid 2^m - 1$ is odd and (i) already forces t odd. The window is the one-line observation: monomial substitution scales tail and degree by the *same* factor,

$$f(x^t) - x^{mt} = (f - x^m)(x^t), \qquad \deg\big(f(x^t) - x^{mt}\big) = t \deg(f - x^m) \leq t \lfloor m/2 \rfloor \leq \lfloor mt/2 \rfloor,$$

the last inequality for every $t \geq 1$ (equality when m is even; for m odd it reads $t(m-1)/2 \leq (mt-1)/2$). Machine-checked for all $m < 200$ and all odd $t < 200$. ☐

Credit. Nothing here is new mathematics. The irreducibility half is a textbook criterion; the contribution is the observation that the window costs nothing, so the criterion transports Kaser–Lemire from a seed degree to all its admissible multiples. The $m=2$ case is the cyclotomic family $\Phi_{3^{k+1}} = x^{2\cdot 3^k} + x^{3^k} + 1$, $n = 2 \cdot 3^k$, which Jeřábek gave on MathOverflow in November 2011 — the same thread in which the conjecture was posed — and which in one direction reproves the classical trinomial fact that $x^{2k} + x^k + 1$ is irreducible over GF(2) if and only if k is a power of 3 (Fredricksen–Wisniewski [9]; see also Golomb–Lee [10]).

Operational criterion. Hypotheses (i)–(ii) collapse to: for every prime $p \mid t$, $v_p(e) = v_p(2^m-1) \geq 1$, i.e. $p \mid 2^m - 1$ and $x^{(2^m-1)/p} \neq 1$ in $\mathbb{F}_2[x]/(f)$. This needs *no* factorization of 2^m-1 — one modular exponentiation per candidate prime — and the candidate primes are found from $\mathrm{ord}_p(2) \mid m$. Write $A(f)$ for the resulting set of *admissible primes*; Theorem 3.3 applies to exactly those t all of whose prime factors lie in $A(f)$. If f is primitive, $A(f)$ is the full set of prime divisors of 2^m-1.

Coverage and its honest reading. Applied to a ledger L of known seeds — here, one certified in-window irreducible for every $m \leq 3000$ plus every in-window irreducible found by a bounded per-degree search — the theorem proves the conjecture on $S(L,N) = \{mt \leq N : m \in L,\ t \geq 2,\ \mathrm{rad}(t) \subseteq A(f) \text{ for one seed } f \text{ of degree } m\}$.

N	$\#(S \cap [1,N])$	$\#S/N$	composites $\le N$	fraction of composites
10^2	22	0.2200	74	0.297
10^3	243	0.2430	831	0.2924
10^4	2086	0.2086	8770	0.2379
10^5	8394	0.0839	90407	0.0929

Table 2: Exact coverage of Theorem 3.3 applied to a ledger of certified in-window seeds of every degree $m \le 3000$ (**FINITE**; `data/composition-coverage.txt`). The apparent health of the first three rows and the drop in the fourth are both artefacts of the ledger, not densities: see the discussion of (B4) in §4.3.

Note the quantifier: the primes of a single t must all be admissible for *one* seed. Table 2 gives the exact counts; 411 members are re-verified by a direct Rabin certificate on $f(x^t)$ (degrees 6 to 20000) and 220 of those a third time by python-flint. The smallest composite *not* in S is 4; the smallest odd composite not in S is 9. No prime and no power of two is ever in S. Asymptotically, $\max_f |A(f)| = 16$ over this ledger (first attained at $m = 360$), so $\#S(L,N) = O((\log N)^{16})$ and the density is zero — but the implied bound evaluates to 6.5×10^7 at $N = 10^5$, so it exceeds N over the whole computed range: **density zero here is proved asymptotically and not observed**.

And the truth is far stronger. Theorem 3.3 produces witnesses whose tail sits at exactly $t \deg(f - x^m)$, i.e. at the top of the window, whereas the minimal subdegree in Arndt's table is ≤ 10 for every $n \le 400$ and tracks $\sim \log_2 n$. The construction is nowhere near the observed behaviour; the analytic ceiling $(n/2 - \log_2 n)$ matches it much better.

3.4 What the sieve proves

Write $h = \lfloor n/2 \rfloor + 1$, so $|W_n| = 2^h$ and $\ell = n - h$. All of the following is **PROVED** (note 13), machine-checked by `lemire_sieve_face.py` with six mutation controls, each shown to trip.

Lemma 3.4 (exact Type I)**.** *Let d be monic of degree k, let $d^* = x^k d(1/x)$ and $c = (d^*)^{-1} \bmod x^{\ell+1}$. Then*

$$A_d := \#\{F \in W_n : d \mid F\} = 2^{h-k} \quad (0 \le k \le h), \qquad A_d = \mathbf{1}[\deg c \le n-k] \quad (h < k \le n).$$

In particular the remainder $r_d = A_d - |W_n|/|d|$ is identically zero *for every monic d of degree $\le h$, so W_n has level of distribution $D = |W_n| = 2^h$ with no error term at all.*

Proof. $F = dm$ with m monic of degree $n-k$, and reversal is multiplicative: $F^* = d^* m^*$. The window condition is $F^* \equiv 1 \bmod x^{\ell+1}$, i.e. $m^* \equiv (d^*)^{-1} \bmod x^{\ell+1}$. As m runs over monics of degree $n-k$, m^* runs over all polynomials of degree $\le n-k$ with constant term 1, so $n-k$ free coefficients are prescribed in positions $1, \dots, \ell$; if $k \le h$ then $h-k$ remain free. □

This is strictly better than the integer analogue, where $[x, x+\sqrt{x}]$ has $|r_d| \le 1$ and the level is $\sqrt{x}/\log x$. It is also *best possible.*

Corollary 3.5 (no level beyond $|W_n|$, not even on average)**.** *For $k > h$, $\sum_{\deg d = k} |r_d| = 2(2^h - 2^{2h-k}) \to 2|W_n|$. So there is no Bombieri–Vinogradov-style averaged level past 2^h either: the total remainder at any single level above h is twice the size of the whole set.*

Corollary 3.6 (the sieve sees only "an interval")**.** *For* every *monic A_0 of degree n, the interval $I(A_0) = \{F :$* $\deg(F - A_0) < h\}$ *has $\#\{F \in I(A_0) : d \mid F\} = 2^{h-k}$ exactly, for every monic d of degree $k \le h$.*

The sieve axioms hold with dimension $\kappa = 1$; Mertens over $\mathbb{F}_q[t]$ is exact ($\log(1/V(y)) = \sum_{N \le y} 1/N + R_y$ with $|R_y| \le 4q^{-y/2}$, so $V(y) = e^{-\gamma}/y\,(1 + O(1/y))$ with no q-dependent constant), and the Jurkat–Richert linear sieve [31] applies with the remainder sum identically zero. The sifting parameter at the prime level is $s = h/(n/2) = 1 + 2/n \to 1$, which is the parity line.

Theorem 3.7 (what the linear sieve gives)**.**

(a) (P_4) For every $\epsilon > 0$ and $n \ge n_0(\epsilon)$, W_n contains an F all of whose irreducible factors have degree $> (\frac{1}{4} - \epsilon)n$; four such factors would overflow the degree, so $\Omega(F) \le 4$.

(b) (P_3, Kuhn weights) For $1/8 \le \alpha < 1/6$ and $n \ge n_0(\alpha)$, W_n contains an F with $\Omega(F) \le 3$ all of whose irreducible factors have degree $> \alpha n$. The margin is $G(\alpha) = 2e^\gamma \alpha \log\big((1-2\alpha)/(4\alpha)\big)$, positive exactly for $\alpha < 1/6$.

(c) *(explicit, no black box) By Bonferroni truncation of the Legendre sieve — legitimate because every divisor occurring has degree* $\leq (2r+1)y \leq h$*, so by Lemma 3.4* every *term is exact —* W_n *contains an element with no irreducible factor of degree* ≤ 3 *for every* $n \geq 28$*, none of degree* ≤ 10 *for* $n \geq 138$*, none of degree* ≤ 30 *for* $n \geq 658$*.*

Two caveats stated plainly. The diary's "$s > 2$ hence at most 3 factors" is off by one: $s > 2$ at $y = (1/4 - \epsilon)n$ gives P_4, and P_3 needs a weight. And n_0 in (a)–(b) is explicit only *modulo one absolute constant*: the Jurkat–Richert error $K(\log D)^{-1/3}$, for which we know of no published value of K. At $\alpha = 1/8$ the computation gives $n_0(1/8) = \max(300, 2825K^3)$, i.e. 354 if $K = 1/2$ and 2828 if $K = 1$.

Theorem 3.8 (exact Brun–Titchmarsh, no error term)**.** *Let* $L = \lfloor h/2 \rfloor$ *and* $G_L = \sum_{d \text{ squarefree, } \deg d \leq L} \prod_{P|d}(|P| - 1)^{-1}$*. Then* $\#\{F \in W_n \text{ irreducible}\} \leq |W_n|/G_L$*.*

Selberg's Λ^2 sieve with weights supported on $\deg d \leq L$ has $\deg[d_1, d_2] \leq 2L \leq h$, so by Lemma 3.4 every entry of the quadratic form is exactly $2^{h-\deg[d_1,d_2]}$ and the diagonalisation leaves no remainder. Since $G_L = L + 1.38 + o(1)$, the bound is $(4 + o(1))|W_n|/n$ — the classical constant — and it is measured at 3.0 to 3.3 times the truth for $n \leq 44$ (**FINITE**). The level restriction $2L \leq h$ is load-bearing: the mutation control that triples L makes the "bound" drop below the truth already at $n = 10$.

3.5 Large q: two thresholds, and what they do not cover

Theorem 3.9 (Hayes/Weil, sharp form: Hsu 1996 = Cohen 2005)**.** *The number of monic irreducibles of degree n over* $\mathbb{F}_q$ *with the top l coefficients prescribed is at least* $q^{n-l}/n - (l+1)q^{n/2}/n$*, positive exactly when* $l < n/2 - \log_q(l+1)$*.*

At $q = 2$ this prescribes the top $n/2 - \log_2 n$ coefficients unconditionally, against the $\lceil n/2 \rceil - 1$ that Kaser–Lemire asks: shortfalls 27 vs 31 at $n = 64$, 503 vs 511 at $n = 1024$, 2037 vs 2047 at $n = 4096$. The same inequality makes Kaser–Lemire a *theorem* for $q > n/2$ (even n) and $q > (n+1)^2/4$ (odd n) — an n-dependent threshold, so it settles no fixed field for large n.

Theorem 3.10 (Bagshaw, arXiv:2401.10399 Cor. 2.5, via the reversal duality)**.** *Let p be an* **odd** *prime and* $q = p^l$ *with* $q > 961e^2p^2 = 7100.883\ldots p^2$*. Then there is* $n_0 = n_0(q)$ *such that for every* $n \geq n_0$ *there is a monic irreducible* $f \in \mathbb{F}_q[T]$ *of degree n with* $\deg(f - T^n) \leq \lfloor n/2 \rfloor$*.*

This is the first *n-independent* q-criterion for the full half-degree window; every earlier one (Hayes/Weil, Hsu, Cohen, Pollack, Ha) has a threshold growing with n. Its input is a level of distribution $\omega < 1/2 + 1/62$ for von Mangoldt in progressions to an *arbitrary* modulus — crucially including T^r, the maximally non-squarefree one, which is why Sawin–Shusterman [21] and Sawin's Acta theorem, both restricted to squarefree moduli, do not apply. The statement is for an individual modulus, not a Bombieri–Vinogradov average.

Its scope, precisely (note 16, `lemire_largeq.py`). Three restrictions, each verified against the source.

(1) p **must be odd, by mechanism and not only by hypothesis.** The paper fixes "an odd prime power $q = p^l$" in its standing notation, and the one lemma the deduction cannot do without — Sawin–Shusterman's Möbius-in-progressions bound — lives in a section opening "From now on, we will assume that the characteristic p of $\mathbb{F}_q$ is odd. Because of this, $\mathbb{F}_q^\times$ admits a unique quadratic character." The whole cancellation runs through Jacobi symbols and the quadratic character of a resultant. *There is no $p = 2$ case.*

(2) **The threshold forces $l \geq 3$.** $q > 7100.883p^2$ with $q = p^l$ is $p^{l-2} > 7100.883$. No prime field and no $q = p^2$ qualifies at any size. The smallest admissible q is $3^{11} = 177147$; the admissible set has $O(X^{1/3})$ members below X. (Minimal admissible q: 3^{11}, 5^8, 7^7, 11^6, 13^6, 23^5, 89^4, 7103^3.) An external control on the enumeration rule: Bagshaw's own remark that improving the Sawin–Shusterman twin-prime constant 685090 to 181157 newly covers exactly $3^{14}, 5^{10}, 13^7, 23^6, 59^5, 61^5, 67^5, 71^5, 73^5, 79^5, 83^5$ is reproduced verbatim by the rule "p^l admissible iff p odd and $p^{l-2} > C$".

(3) $n_0(q)$ **is ineffective**, so it cannot be combined with a finite computation to close even one admissible q.

Hence the correct reframing is narrow: *for a sparse infinite set of q, Kaser–Lemire holds for all large n, with an n-independent hypothesis on q.* It does not touch characteristic two, prime fields, or $q = p^2$. The sentence "the residual problem is small q" is false, and it was asserted in an earlier draft of note 15 before note 16 checked the source (§8).

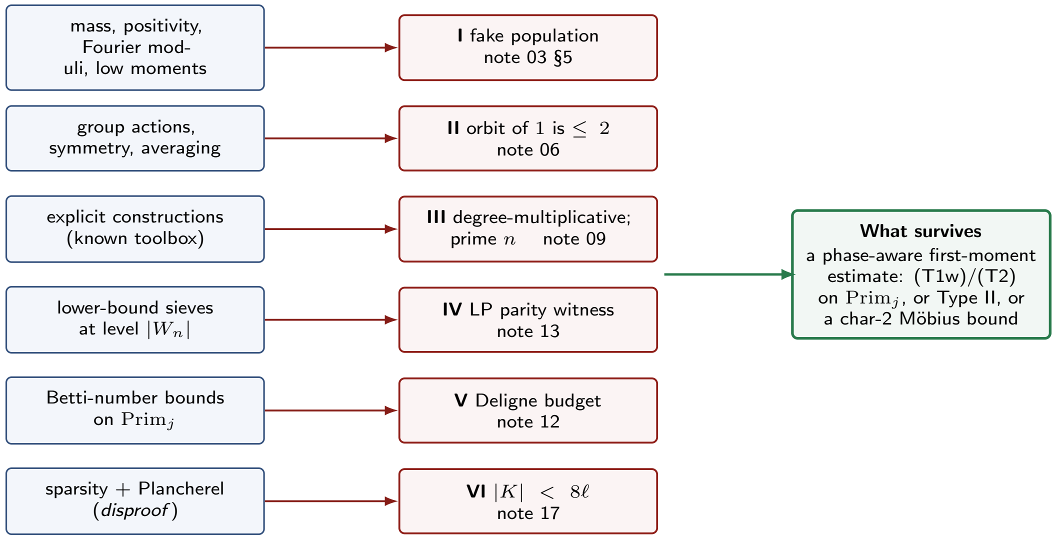

Figure 2: Which class of argument each barrier eliminates. Every barrier is a statement about *methods*, not about the truth of the conjecture. The green box is Section 7.

4 Barriers

This is the heart of the document. Six barriers are stated, each with a proof or proof sketch, an exact scope, and a statement of which class of argument it eliminates. None of them is an independence result: the conjecture is almost certainly true, and in every case a genuinely different input is not blocked. What they do is close, permanently, six directions that look promising from the outside.

4.1 Barrier I (moduli-only): an explicit fake population

Proposition 4.1 (**PROVED**; note 03 §5, note 04). *Fix ℓ, n, $a = \ell - \lceil \log_2 \ell \rceil - 1$ and $K = \ker(\mathcal{E}_\ell \to \mathcal{E}_{a-1})$. Let $m(g) = N_{a-1}(\text{class of } g)/|K|$ be the cylinder-mean population, put $c = m(1)/(1 - 2^{a-1-\ell})$ and*

$$F(g) \;=\; m(g) \;+\; c\big(2^{\,a-1-\ell}\mathbf{1}_K(g) - \delta_{g,1}\big).$$

Then $F \geq 0$ on $\mathcal{E}_\ell$, $F(1) = 0$, $\sum_g F(g) = 2^n$, $\widehat{F}(\chi)$ equals the true $\widehat{N}(\chi)$ for every χ of conductor $< a$, and $|\widehat{F}(\chi)| \approx 2^{n-\ell} \leq (a-1)2^{\lceil n/2 \rceil}$ — inside the Weil bound for its conductor by a factor about a — for every χ of conductor $\geq a$. Its second moment is below *that of N.*

Consequently: **no argument whose only inputs are total mass, nonnegativity, the exact low-conductor populations, per-character Weil bounds on the Fourier moduli, and bounds on low moments can prove (REL)**, let alone $N_\ell(1) > 0$ — because every such hypothesis holds for F, and $F(1) = 0$. The second-moment ratios $\|F\|_2^2/\|N\|_2^2$ were measured at $0.22,\ 0.15,\ 0.12$ at $(\ell, n) = (12, 25), (16, 33), (20, 41)$, with the construction verified numerically ($\min F = 0,\ F(1) = 0$).

Scope, and what it kills. It kills every route through Theorem 3.2: sharpening $\varepsilon(\ell)$, or even proving the Sato–Tate second moment, applies verbatim to F. It is the population-level form of a second obstruction found independently on the L-function side: $Q(T) = 1 - 2^{(n+1)/2}T^n + 2^nT^{2n}$ is an integral Weil polynomial with $p_m(Q) = 0$ for all $m < n$ and $p_n(Q) = n2^{(n+1)/2}$, so multiplying any admissible L-polynomial product by powers of Q preserves degree, integrality, RH, the functional equation, the 2-adic Newton polygon and every power sum below n, while driving $|p_n|$ to about 0.7 of its trivial bound (note 02 §5B). *No generic invariant of these L-functions can supply the saving.* A proof must use that N is a Λ-weighted prime count beyond its Fourier moduli — that is, phase information across the family.

What it does not kill. A class-specific lower bound reading the description "all odd-power Galois-ring traces vanish", and any phase-aware correlation among the $S_n(\chi)$. Barrier I is a barrier for a class of methods; it is not an independence result.

4.2 Barrier II (symmetry): the orbit of the identity has size at most two

Proposition 4.2 (**PROVED**; note 06, `lemire_adams_check.py` and the symmetry scans)**.** *Let φ be a degree-preserving substitution symmetry of the degree-n irreducibles over $\mathbb{F}_2$. Then $\varphi \in \mathrm{PGL}_2(\mathbb{F}_2)$, and φ induces a permutation of $\mathcal{E}_\ell$ commuting with N_ℓ if and only if φ fixes the place at infinity, i.e. lies in the Borel subgroup $B(\mathbb{F}_2) = \{\mathrm{id},\ x \mapsto x+1\} \cong \mathbb{Z}/2$. Adjoining the Galois/Adams action $g \mapsto g^\alpha$ (α odd, which fixes 1) does not enlarge orbits. Hence the orbit of the identity class under the full group of structural symmetries has size at most $2 < 4\ell^2$. The Hecke action of the ray class group is transitive on $\mathcal{E}_\ell$ but shifts the degree by $\deg L$ and fixes no degree-n fibre.*

Therefore no group action can place the identity class in the non-exceptional set of Theorem 3.2, and none can prove $N_\ell(1) > 0$.

What was tested (exact, $3 \le \ell \le 8$). All six elements of $\mathrm{PGL}_2(\mathbb{F}_2)$ preserve degree and irreducibility (checked to $\#\mathrm{irr} = 7710$); only $\{\mathrm{id}, x \mapsto x+1\}$ descends, the four elements with $c \neq 0$ failing the block condition on all 2^ℓ classes because they read low coefficients that $\langle f \rangle_\ell$ does not see. Power maps $\alpha \mapsto \alpha^k$ with $\gcd(k, 2^n - 1) = 1$ and k not a power of 2 — the largest transitive group on the primes — are genuine degree-n bijections but none descends: every class conflicts, for all twelve exponents tested at $(4,9), (5,11), (6,13)$. Translation is an involution with $\sigma(1) = \langle (1+x)^n \rangle_\ell$, which equals 1 only when $\binom{n}{k}$ is even for all $1 \le k \le \ell$; measured $\sigma(1) = 15, 51, 255, 3$ at $(5,11), (6,13), (7,15), (8,17)$ and $85, 1$ at $(6,14), (7,16)$. (This corrects an earlier claim in these notes that translation fixes the identity class; the conclusion is unchanged, since the orbit is $\{1, \sigma(1)\}$ either way.)

The Gorodetsky–Kovaleva symmetry is the monomial case of this action

Gorodetsky and Kovaleva [14] prove an exact fixed-q identity that replaces Weil's conductor factor k by $\gcd(k, q^n - 1)$: for $\chi_{k,\psi}(f) = \psi(p_{-k}(f))$ a primitive character modulo T^{k+1} and $k' = \gcd(k, q^n - 1)$,

$$\sum_{f \in M_{n,q}} \Lambda(f)\chi_{k,\psi}(f) = \sum_{f \in M_{n,q}} \Lambda(f)\chi_{k',\psi}(f), \quad \text{hence} \quad \Big|\sum_f \Lambda(f)\chi_{k,\psi}(f)\Big| \le q^{n/2}\gcd(k, q^n - 1).$$

This is a genuine, unconditional, fixed-q improvement over Weil for our family, proved in three lines with no geometry, and it is worth knowing that such a thing exists at all. **But it is exactly the power/Adams action of Proposition 4.2**, and it is strong there only because the phase is a *single* monomial γ^{-k}, which the power map $x \mapsto x^{\gcd(k,q^n-1)}$ collapses wholesale. A general exact-conductor-j character has phase $\psi(\mathrm{Tr}P(\gamma))$ with P supported on all odd $k \le j$; the power map moves the whole tuple at once, and Proposition 4.2 measured the resulting orbit.

The counting is decisive. A single-position ("monomial phase") character is trivial on all but one factor of $\mathcal{E}_j = \prod_{k \text{ odd} \le j} \mathbb{Z}/2^{e_k}$, so their number is $\sum_k (2^{e_k} - 1) + 1 \sim 2j \ln j$ against 2^j characters in the dual:

j	#single-position	fraction of the dual
24	69	$2^{-17.9}$
100	377	$2^{-91.4}$
200	853	$2^{-190.3}$
1024	6145	$2^{-1011.4}$

A layer $X_{j,s}$ has about 2^{j-1} characters. Bounding a 2^{-1011} fraction of them *perfectly* — $|S_n(\chi)| = 0$ — and the rest by Weil leaves the layer sum exactly where Weil left it. So the identity cannot bound $T_{j,s}$, and cannot give (HWO). (Note 15 §4. An earlier version of note 15 listed this as the sweep's top lever, "short of (HWO) by the absolute constant 4 rather than by a factor ℓ"; that statement is correct per character and vacuous per layer, and it was withdrawn on review — see §8.)

4.3 Barrier III (construction): the toolbox is degree-multiplicative

Fix the *provable toolbox* T: monomial composition; general composition $f \mapsto f(\sigma)$ with $\deg \sigma = k$; the Meyn R-transform and the Q-transform $f \mapsto f(x^2 + x)$; Möbius maps; Brawley–Carlitz composed products; Cohen/Kyuregyan recursive $f \mapsto h^m f(g/h)$; cyclotomic Φ_M and its factors; Carlitz–Drinfeld cyclotomic; and the norm from $\mathbb{F}_{2^k}[x]$.

Proposition 4.3 ((B0) degree-multiplicativity, **PROVED**). *Every row of T produces a degree that is a proper multiple of a smaller input degree, or is degree-preserving, or is $\phi(M)$ / $\mathrm{ord}_M(2)$ for a cyclotomic modulus. Structurally: a fixed algebraic recipe ρ of degree e applied to a root α of a degree-n irreducible gives $\beta = \rho(\alpha)$ with $\mathbb{F}_2(\beta) \subseteq \mathbb{F}_{2^{ne}}$, so $[\mathbb{F}_2(\beta) : \mathbb{F}_2] \mid ne$.*

Proposition 4.4 ((B1) prime-n blocker, **PROVED**). *Let $n \geq 5$ be prime. No construction in T produces an irreducible of degree n from data of smaller degree.*

Proof sketch, row by row. Monomial and general composition, composed products, the recursive $h^m f(g/h)$ and the norm all give degree mk with $k \geq 2$; n prime forces $m = 1$, and a degree-1 seed has order 1, so hypothesis (i) of Theorem 3.3 fails for every $t \geq 2$ and the other rows return the substitution itself. The R- and Q-transforms and the norm at $k = 2$ give even degree. Möbius maps preserve degree, and their orbit is the size-≤ 2 orbit of Barrier II. For cyclotomic Φ_M, $\phi(M)$ is prime only for $\phi(M) = 2$. For an irreducible *factor* of Φ_M of degree $\mathrm{ord}_M(2) = n$: every irreducible divides $\Phi_{\mathrm{ord(root)}}$, so this is a tautology rather than a construction — and for prime n the relevant modulus is a large generic divisor of $2^n - 1$, carrying no small-conductor structure and no information about the window. Carlitz–Euclid overshoots to degree $\sim 2^\ell \deg F$. □

Proposition 4.5 ((B2) power-of-two blocker; (B3) lacunarity, **PROVED**). *$n = 2^a$ is never reached by monomial composition, since $e \mid 2^m - 1$ is odd and (i) forces t odd. For a seed f of degree m with admissible primes $A(f)$, $\#\{t \leq X : \mathrm{rad}(t) \subseteq A(f)\} \leq \prod_{p \in A(f)}(1 + \log X/\log p)$, so for a* fixed finite *ledger L, $\#S(L, N) = O((\log N)^W)$ with $W = \max_f |A(f)|$: density zero.*

(B4) What is only observed, and why it is not a density. Inside the range where the ledger itself grows — every composite $n \leq 6000$ has all its proper divisors among the certified seeds — the measured coverage is a substantial fraction (Table 2), and the drop at 10^5 is the ledger running out, not a density appearing. Both readings are wrong, in opposite directions. With $W = 16$ the bound of (B3) evaluates to 6.5×10^7 at $N = 10^5$: vacuous over the entire computed range. The temptation to read the table as positive density is precisely the circularity the barrier is about: "a positive proportion of m carry an in-window seed" *is* Kaser–Lemire. Theorem 3.3 multiplies a known answer; it does not produce one.

(B5) The reduction, and honest scope. Combining (B0)–(B3), the construction angle reduces the conjecture to the degrees it cannot reach — the primes, the powers of two, and those composites none of whose proper divisors carries a suitable seed — and stops. Barrier III rules out the *known* toolbox as listed; it is **not** a logical impossibility theorem. A genuinely new construction producing a prime degree from smaller data, or a seed family whose degrees run in an arithmetic progression, would evade it. No such construction is known. What has been gained relative to a vaguer formulation is that the obstruction is now *located*: at prime n, where every row is silent for the structural reason that $n = mk$ admits no factorization with both factors ≥ 2.

The literature ceiling, corrected. Three different quantities get called "the prescribed-coefficient ceiling" and they differ by more than a constant.

- **Top positions, any q — the relevant one.** Hayes plus Weil in the sharp form of Theorem 3.9: $n/2 - \log_q(l+1)$, i.e. $n/2 - \log_2 n$ at $q = 2$. *The entire gap is $\sim \log_2 n$ coefficients.*
- **Arbitrary positions, fixed q.** Pollack [27]: $\lfloor (1-\epsilon)\sqrt{n} \rfloor$ coefficients in arbitrary positions, uniformly in q. Ha [28]: $(1/4 - \epsilon)n$ arbitrary positions but only for $q \geq q_0(\epsilon)$; at arbitrary q his Theorem 1.3 gives $r \leq n/10$ for $n \geq 52$, i.e. $n/10$ at $\mathbb{F}_2$. Garefalakis: $(1/3 - \epsilon)n$ *consecutive* coefficients set to zero, in any position.

An earlier version of note 09 quoted the $\sqrt{n}$ figure as the ceiling for this problem. That is wrong by a power: $\sqrt{n}$ is the arbitrary-position ceiling, and the ceiling for the top positions is $n/2 - \log_2 n$. The correction matters in the direction of *more* optimism about the analytic route: the required saving over Weil is logarithmic in X, not a power.

Finally, these are counting theorems and not constructions, so Barrier III does not touch them. The explicit route stops at multiples of known degrees; the analytic route stops $\log_2 n$ short. The two stop for different reasons and neither is repaired by enlarging the other.

n	h	$\mathrm{LP}(n,h{-}1)$	$\mathrm{LP}(n,h)$	$\mathrm{LP}(n,h{+}1)$	$\mathrm{LP}(n,h{+}2)$	I_n	$k_{\max}(n)$	$k_{\max}-h$
6	4	0	4	8	9	9	4	0
7	4	0	8	16	18	18	4	0
8	5	0	8	20	26	30	5	0
9	5	0	16	40	52	56	5	0
10	6	0	0	48	76	99	7	1
11	6	0	0	96	144	186	7	1
12	7	0	0	96	192	335	8	1
13	7	0	0	192	336.23	630	8	1
14	8	0	0	32	344.08	1161	9	1
15	8	—	0	64	—	2182	9	1
16	9	—	0	0	298.08	4080	11	2
17	9	—	0	0	?	7710	≥ 11	≥ 2

Table 3: The level at which the divisor data forces a prime (**FINITE**; note 13, `data/sieve-lp-levels.txt`; values in the M_n normalisation, so divide by 2^{n-h} for the W_n scale). $k_{\max}(n) = \min\{K : \mathrm{LP}(n, K) > 0\}$. The barrier switches on at $n = 10$: for $6 \leq n \leq 9$ the exact data at level 2^h *does* force a prime, so small-n intuition here is actively misleading. $\mathrm{LP}(n, n) = I_n$ always, and LP is nondecreasing in K; both checked.

4.4 Barrier IV (parity/sieve): an exact prime-free population

Every lower-bound sieve — Brun, Selberg-Λ^2-plus-Buchstab, Rosser–Iwaniec, Kuhn or Richert weights, and any linear combination — only ever learns the numbers A_d. Formally, a *lower-bound sieve certificate at level* K is a family $(\lambda_d)_{\deg d \leq K}$ with $\sum_{d|F,\ \deg d \leq K} \lambda_d \leq \mathbf{1}[F \text{ irreducible}]$ for every monic F of degree n; it yields $\#\{\text{irred. in } W_n\} \geq \sum_d \lambda_d A_d$. The constraint is required on all of M_n and not merely on W_n, because a sieve never learns which subset of M_n its sequence is.

Theorem 4.6 (LP duality, **PROVED**; note 13 Thm. 10)**.** *Put*

$$\mathrm{LP}(n, K) = \min\Big\{ \sum_{F \text{ irred.}} w(F) \ :\ w \geq 0 \text{ on } M_n,\ \sum_{F:\, d|F} w(F) = 2^{\,n-\deg d} \text{ for all } \deg d \leq K \Big\}.$$

Then $2^{h-n}\mathrm{LP}(n, K)$ is exactly the supremum of $\sum_d \lambda_d A_d$ over certificates at level K. In particular $\mathrm{LP}(n, K) = 0$ if and only if a nonnegative w on M_n exists that vanishes on every irreducible and has the exact Type-I data — and then no lower-bound sieve at level 2^K can prove that W_n contains an irreducible.

Theorem 4.7 (the parity barrier, with an exact rational witness; **PROVED**+**FINITE**)**.** *For each $n \in \{10, 11, 12, 13, 14, 15\}$ there is an explicit nonnegative rational w on M_n with $w(F) = 0$ for every irreducible F of degree n and $\sum_{F:\, d|F} w(F) = 2^{\,n-\deg d}$ exactly, for every monic d of degree $\leq h$. Consequently $\mathrm{LP}(n, h) = 0$: **no lower-bound sieve using only the Type-I data of W_n at its own exact level can prove that W_n contains an irreducible.***

The witnesses are committed one exact fraction per polynomial (supports 113, 122, 237, 242, 482, 501); the checker re-verifies over $\mathbb{Q}$ with exact rational arithmetic — nonnegativity, vanishing on every irreducible, and all $2^{h+1} - 1$ Type-I equalities — and two mutation controls (moving a unit of mass onto an irreducible; perturbing one value by 1/7) both trip. The support comes from an LP vertex, but the values are re-solved exactly over $\mathbb{Q}$ on that support, so no floating-point result is load-bearing.

Remark 4.8 (the textbook example is not exact enough)**.** The classical parity population $w = 1 + \lambda$ (λ the Liouville function) has $\sum_{F:d|F} w(F) = 2^{n-k} + \lambda(d)L(n-k)$ with $L(j) = 2^{j/2}$ (j even), $-2^{(j+1)/2}$ (j odd) — *never* zero. Its Type-I remainders are of exact square-root size, which the window's identically zero remainders distinguish. The LP says more: demanding the support lie inside $\{\Omega \text{ even}\}$ is infeasible for every $n \in \{8, \ldots, 16\}$ tested, with a forced odd-Ω mass of square-root order (fraction $0.1563, 0.1108, 0.1406, 0.0423, 0.0729, 0.0321, 0.0375, 0.0094$ at $n = 8, 10, \ldots, 16$). And if $w = 1 + \theta$ with θ completely multiplicative and $\theta(P) = -1$ on irreducibles then $\theta = \lambda$; so an exact prime-free population is necessarily non-multiplicative, which is why the construction is an LP witness and not a formula.

Two readings of Table 3 are worth stating. For $10 \leq n \leq 15$ the first sufficient level is $k_{\max}(n) = h + 1$: *one* more level of exact data would settle the window — and by Corollary 3.5 that level is not available even in principle. At $n = 16$ even $h + 1$ is insufficient, and $k_{\max} - h$ is rising, consistent with the asymptotic expectation $k_{\max}(n) \sim n$ (the parity principle is $s > 2$, i.e. $K > n$). Moreover the values at $h + 1$, put on the W_n scale, are far below the truth ($48 \cdot 2^{-4} = 3$ against $I_{10}(1) = 7$; $32 \cdot 2^{-6} = 0.5$ against 19 at $n = 14$): even a sieve handed one impossible extra level would prove only a fraction of the truth.

Proposition 4.9 (a sieve proof would prove Legendre for $\mathbb{F}_2[t]$; PROVED)**.** *For every monic A_0 of degree n and every certificate at level $K \leq h$, $\sum_d \lambda_d A_d \leq \#\{F \text{ irreducible} : \deg(F - A_0) < h\}$. Hence the best level-$2^h$ sieve lower bound for W_n is at most the* minimum over all 2^ℓ top-half patterns *of the irreducible count in that window.*

Proof. Sum the certificate inequality over $F \in I(A_0)$ and use Corollary 3.6: the left side is $\sum_d \lambda_d 2^{h - \deg d} = \sum_d \lambda_d A_d$. ☐

So a sieve proof of Kaser–Lemire at the window's own level would *simultaneously* prove Legendre's conjecture for $\mathbb{F}_2[t]$ — every window $\{A_0 + g : \deg g \leq \lfloor n/2 \rfloor\}$ contains an irreducible. **The sieve face cannot separate the identity class from the uniform statement**; it is blind to the very distinction Barriers I–III are about. (Conversely a single prime-free window at some n would kill the level-2^h route outright; none exists for $n \leq 16$, where the minimum over windows is $2, 2, 2, 2, 5, 4, 4, 6, 13, 11, 24$.)

Where the extra input must come from. Iwaniec's well-factorable form replaces $\sum_{\deg d \leq K} |r_d|$ by $\sum_d \lambda_d r_d$ with λ well-factorable. For $\deg d = k > h$ we have $r_d = A_d - 2^{h-k}$ with $A_d \in \{0, 1\}$, so $\sum_d \lambda_d r_d$ is, up to a main term, a bilinear count of products landing in the window. *The well-factorable remainder sums above level h ARE the missing Type II estimate*, not an alternative to it. And Type II lands back on the same characters:

Proposition 4.10 (Type II transplant, PROVED; note 13 Prop. 13)**.** *Let $M + L = n$ and let α, β be arbitrary coefficients on the monics of degrees M, L. Then*

$$\sum_{\deg m = M} \sum_{\deg l = L} \alpha_m \beta_l \, \mathbf{1}[ml \in W_n] \;=\; 2^{-\ell} \sum_{\chi} A_M(\chi) B_L(\chi),$$

with $A_M(\chi) = \sum_m \alpha_m \chi(\langle m \rangle_\ell)$ and similarly B_L.

So every Type II sum over the window is a χ-average over the *same* Hayes family that carries $S_n(\chi)$, with $A_M(\chi) B_L(\chi)$ in place of $S_n(\chi)$: the sieve route does not bypass (REL)/(HWO), it re-weights the same sum over the same family. Stated honestly in both directions: Type II gives the count and more (Möbius cancellation in the window, uniformly in the coefficients); (HWO) gives the count and not the bilinear forms. Neither implies the other formally, and Type II is strictly stronger in content.

4.5 Barrier V (Deligne budget): a Betti bound alone can never suffice

By Katz [11], the exact-conductor-j characters are the $\mathbb{F}_q$-points of a smooth affine $\mathbb{F}_2$-scheme Prim_j of dimension j, carrying a lisse L_{univ} of rank $j - 1$, pure of weight one, with $\det(1 - T \,\mathrm{Frob}_\chi \mid L_{\mathrm{univ}}) = L(\chi, T)$. Writing Ξ_n for the n-th Adams operation, $\Xi_n(L_{\mathrm{univ}})$ is a virtual lisse sheaf pure of weight n with $\mathrm{Tr}(\mathrm{Frob}_\chi \mid \Xi_n L_{\mathrm{univ}}) = -S_n(\chi)$, and the conductor-$j$ sum is a Lefschetz trace.

Proposition 4.11 (the budget; PROVED, note 12 Prop. 1)**.** *Let $X/\mathbb{F}_2$ be separated of finite type, $\mathcal{F}$ a virtual lisse sheaf pure of weight n with $|\mathrm{Tr}(\mathrm{Frob}_x \mid \mathcal{F})| \leq R(\#k(x))^{n/2}$ at every closed point, $A_r = \sum_{x \in X(\mathbb{F}_{2^r})} \mathrm{Tr}(\mathrm{Frob}_x \mid \mathcal{F})$, $C = \sum_i h_c^i(X \otimes \overline{\mathbb{F}_2}, \mathcal{F})$ and $i_{\max}$ the top non-vanishing degree. Then $|A_r| \leq C \, 2^{r(n + i_{\max})/2}$ (Deligne), against the trivial $|A_r| \leq R \, \#X(\mathbb{F}_{2^r}) \, 2^{rn/2}$. Hence a saving of at least G at $r = 1$ requires*

$$C \;\leq\; \frac{R \, \#X(\mathbb{F}_2)}{G \, 2^{i_{\max}/2}}.$$

For $X = \mathrm{Prim}_j$, $R = j - 1$, $\#\mathrm{Prim}_j(\mathbb{F}_2) = 2^{j-1}$ and $G = 4\ell$ this reads $C \leq (j-1) 2^{j - 1 - i_{\max}/2} / (4\ell)$, i.e. with $k := 2j - i_{\max}$, $k \geq 2 \log_2\big(8\ell C/(j-1)\big)$.

$i_{\max}$	the budget requires	verdict
$2j$ (top)	$C \leq (j-1)/(8\ell) < 1/8$	**impossible**
$2j - 1$	$C \leq (j-1)/(4\sqrt{2}\,\ell) < 0.177$	**impossible**
$2j - 2$	$C \leq (j-1)/(4\ell) < 1/4$	**impossible**
$2j - k$	$2^{k/2} \geq 8\ell C/(j-1)$; with $C = 1$, $j = \ell$: $k \geq 6$	needs $k = \Theta(\log \ell C)$
$j + 1$	$C \leq (j-1) 2^{(j-1)/2}/(8\ell)$	exponential room
j (middle)	$C \leq (j-1) 2^{j/2}/(8\ell)$	exponential room

So the Betti sum is not the binding constraint; the cohomological degree is. If H_c^{2j}, H_c^{2j-1} or H_c^{2j-2} is nonzero, then no bound on C — not even $C = 1$ — gives (HWO) through Deligne. Conversely once the cohomology stops far enough below the top, *any* polynomial-in-j bound on C suffices with exponential margin.

Proposition 4.12 (middle concentration is unavailable; **PROVED** for the geometry, note 12 Prop. 2)**.** $\mathrm{Prim}_j \cong \mathbb{G}_m \times \mathbb{A}^{j-1}$ *as an* $\mathbb{F}_2$*-scheme; hence* $H_c^i = 0$ *for* $i < j$ *by Artin vanishing and duality, so* $j \le i_{\max} \le 2j$*. Moreover* $\mathbb{G}_m$ *acts on* $\mathcal{E}_j$ *over* $\mathbb{F}_2$ *by* $\sigma_t : x \mapsto tx$ *and the trace function of* $\Xi_n(L_{\mathrm{univ}})$ *is* σ*-invariant for every* n*:* $S_n(\chi \circ \sigma_t) = S_n(\chi)$ *and* $N_j(\sigma_t g) = N_j(g)$*. Consequently (under the descent hypothesis of §6)* $i_{\max} \ge j + 1$*: the cohomology is* not *concentrated in the middle degree* j*.*

Proof of the invariance. σ_t is a ring automorphism of $\mathbb{F}_q[x]/x^{j+1}$ fixing the augmentation, hence a group automorphism of $\mathcal{E}_j$ defined over $\mathbb{F}_2$. For monic F of degree n put $F_t(T) = t^n F(T/t)$; then $\sigma_t(\langle F\rangle_j)(x) = (tx)^n F(1/(tx)) = \langle F_t\rangle_j$, and $F \mapsto F_t$ is a Λ-preserving bijection of the monic degree-n polynomials. ☐

This corrected our own published open question. Note 10, which is the document a specialist would have been handed, asked for a uniform bound on the Betti sum $C(2, j, \Xi)$ and asserted that "a polynomial-in-j bound would give (HWO)". Proposition 4.11 shows that implication is *false* unless the degree is also controlled. The question was rewritten as (Q1′), about the *pair* $(i_{\max}, C)$; see §6 and §7.1. This is a barrier against a class of arguments that includes the only published route: Katz's own Theorem 8.2 — the sole polynomial effective-dimension bound for a layer sum, and available only for $p > 2n - 1$ — proceeds by slicing Prim_n into lines and summing trivially over the rest, which is exactly the $i_{\max} = 2\dim - 1$ shape and delivers $\mathrm{Sav} = (j-1)\sqrt{q}/3$, short of 4ℓ by a factor ~ 8.5 at $q = 2$ and unavailable at $p = 2$ in any case.

4.6 Barrier VI: the disproof template fails too

Barriers I–V close routes to a proof. This one closes the only available route to a *disproof* of the sufficient statement (CYL) at fixed q, which is a different and rarer kind of result. The template is Sawin's [19]: a *sparse vanishing locus* plus *Plancherel*, which fixes the total second moment, forces some single value to be enormous — and in his setting this proves that certain fixed-q short-interval sums and moments of Dirichlet L-functions over $\mathbb{F}_q[u]$ to prime-power modulus do *not* admit square-root cancellation.

Let $\mathrm{TM} = \sum_{\psi \in \widehat{K}} |A_\psi|^2$, $\mathrm{NTM} = \mathrm{TM} - A_1^2$, and $M = \max_{\psi \ne 1} |A_\psi|$.

Lemma 4.13 (Plancherel and parity; **PROVED**, note 17)**.** $\mathrm{TM} = |K| \sum_{g \in K} N(g)^2$*,* $A_1 = N_{a-1}(1)$*, and* $\mathrm{NTM} = |K| \sum_{g \in K} (N(g) - A_1/|K|)^2$*. Moreover* $A_\psi \equiv A_1 \pmod 2$ *for every* ψ*; so* ***if*** A_1 ***is odd then no*** A_ψ ***vanishes at all****, for a reason with no equidistribution content.*

Lemma 4.14 (forcing, and its ceiling)**.** *With* $Z_\tau = \#\{\psi \ne 1 : |A_\psi| \le \tau\}$*, if* $Z_\tau \le |K| - 2$ *then* $M^2 \ge (\mathrm{NTM} - Z_\tau \tau^2)/(|K| - 1 - Z_\tau)$*. In particular* $M \le \sqrt{\mathrm{NTM}}$ *unconditionally, so* no vanishing pattern whatever can refute (CYL) at a given ℓ unless $\mathrm{NTM} \ge 2^{2\ell-2}$.

Proposition 4.15 (the template's reach is $\Theta(\ell^{3/2} 2^{-\ell/2})$; **PROVED**)**.** *By* (1)*,* $\sqrt{\mathrm{NTM}} < \sqrt{8\ell} \cdot \mathrm{rms}$*: the forcing gains at most* $\sqrt{8\ell}$ *over the root mean square. Substituting the random-phase model* $\mathbb{E}|A_\psi|^2 = (\mathrm{cond}(\psi) - 1)2^{n-a+1}$ *— whose validity is exactly the measured "aggregate covariance* ≈ 0*" of note 07 — and using that every* K*-bit weight lies in* $[a, \ell]$*,*

$$8\,\ell^{3/2} 2^{-\ell/2} \;\le\; \frac{\sqrt{\mathrm{NTM}_{\mathrm{model}}}}{2^{\ell-1}} \;\le\; 32\,\ell^{3/2} 2^{-\ell/2}.$$

The bracket is asserted in code and verified against the exact $\Sigma = \sum_{\psi \ne 1} (\mathrm{cond}(\psi) - 1)$ for every ℓ in 11..400 at both endpoints. **At $\ell = 200$ the exact model gives reach 3.21×10^{-26} at $n = 401$ and 4.54×10^{-26} at $n = 402$: the template is short of refuting (CYL) by a factor 3×10^{25}.**

Verdict. Both hypotheses of the template fail, independently. Sparsity fails totally: $Z = 0$ at every one of the 26 endpoints, not "small" but zero, and the near-vanishing set has no structure either (the smallest m values form a coset of a subspace of $\widehat{K}$ only for $m = 2$, which is trivial, and once in 26 endpoints for $m = 4$, against a chance rate of 0.8%; and $|A_\psi|$ is not monotone in $\mathrm{cond}(\psi)$). Mass fails, and that is the decisive half: from $\ell = 22$ (odd endpoint) and $\ell = 23$ (even) on, *no* vanishing pattern whatsoever — not even $A_\psi = 0$ for all but one ψ — can force $\max |A_\psi| \ge 2^{\ell-1}$, and by Proposition 4.15 it stays that way for every larger ℓ.

ℓ	n	$\lvert K\rvert$	NTM/model	Z	max /thr	forced/thr	ceiling/thr	refutable?
12	25	64	0.907	0	2.4297	1.0698	8.4909	yes
12	26	64	0.955	0	4.0059	1.5527	12.3239	yes
13	27	64	0.918	0	1.7383	0.7979	6.3334	yes
16	33	64	1.238	0	0.9215	0.3692	2.9306	yes
16	34	64	0.982	0	1.2083	0.4651	3.6915	yes
18	37	128	1.153	0	0.8099	0.2689	3.0299	yes
18	38	128	0.930	0	0.8716	0.3415	3.8484	yes
20	41	128	1.091	0	0.4555	0.1387	1.5627	yes
20	42	128	1.187	0	0.6114	0.2046	2.3060	yes
21	43	128	0.864	0	0.2729	0.0897	1.0105	yes
21	44	128	1.128	0	0.3624	0.1449	1.6329	yes
22	45	128	1.028	0	0.1970	0.0710	0.7997	**NO**
22	46	128	1.051	0	0.3002	0.1014	1.1431	yes
23	47	128	0.964	0	0.1526	0.0498	0.5610	**NO**
23	48	128	1.122	0	0.2340	0.0759	0.8559	**NO**
24	49	128	0.925	0	0.0855	0.0353	0.3976	**NO**
24	50	128	1.091	0	0.1458	0.0542	0.6108	**NO**

Table 4: The Plancherel forcing test (**FINITE**; note 17, `lemire_cylinder_plancherel.py`; 26 endpoints $12 \le \ell \le 24$ computed, a representative subset shown). Threshold $= 2^{\ell-1}$; "ceiling" is $\sqrt{\text{NTM}}$, the template's absolute reach; "refutable" asks whether $\text{NTM} \ge 2^{2\ell-2}$, i.e. whether *any* vanishing pattern could suffice. $Z = 0$ at all 26 endpoints, and at $(23, 47)$ that is a *proof*, since $A_1 = 2148019381$ is odd.

Why, in one sentence. Sawin's argument works because his Plancherel group has about q^{n-1} characters while the relevant sums are supported on $q^{\sim 2n/(p^v+1)}$ of them, so the forcing gain is exponential in the parameter. Here the group is $\mathcal{E}_\ell/\mathcal{E}_{a-1}$, and $a = \ell - \lceil \log_2 \ell \rceil - 1$ is chosen by the Haar telescope, not by us, so only $O(\log \ell)$ coefficients are free. $|K| < 8\ell$ **is a feature of the reduction, and any sparsity/concentration argument on K is capped at $\sqrt{8\ell}$** — which kills the transplant before any data are looked at.

Two by-products. (i) The finite claim in the roadmap note [33] is confirmed at six sizes it had never seen: the odd endpoints satisfy (CYL) from $\ell = 16$ on and the even from $\ell = 18$ on, with every earlier endpoint false — and those earlier failures are now *predicted* rather than anomalous, since the model reach exceeds 1 below $\ell \approx 21$. (ii) A single second-moment inequality, $|K| \sum_{g \in K}(N(g) - A_1/|K|)^2 < 2^{2\ell-2}$, implies (CYL) with no per-character information at all, and it holds exactly at $\ell = 22, 23, 24$. That does not evade Barrier I: the fake population of Proposition 4.1 has $A_\psi(F) = -c$ for every $\psi \ne 1$ with $c \approx 2^{n-\ell}$, so it violates (CYL) outright by a factor 4 to 8. What it records is how much slack (CYL) carries.

The transferable lesson. A sparsity+Plancherel disproof gains at most the square root of the size of the group Plancherel is taken over. Before transplanting one, compare that square root against the ratio between the family's root mean square and the target threshold. Here it is $\sqrt{8\ell}$ against $\Theta(\ell 2^{-\ell/2})$, and the comparison is settled with no arithmetic input at all.

5 Five mechanism shapes, worked to verdicts

Before the barriers were isolated, five candidate solution shapes — the ones that come to mind first — were each worked to a verdict with primary literature and exact computation. All five are negative, each with an exact obstruction, and all five stop at the same place. This section compresses note 04; each paragraph names the obstruction and the number that pins it.

1. A Witt-tower trace formula with a small virtual character (PROVED closed). Hypothesis: $T_{j,s}$ is the trace of Frobenius on a virtual object of dimension polynomial in j. It is not, and for a structural reason: the four Artin–Schreier–Witt quotient covers are *nested*, so the alternating combination $A - B - C + D$ is inclusion–exclusion and equals the layer with multiplicity $+1$ everywhere. There is no negative part; the "virtual" module is the honest H^1_c of the layer, pure of weight one, of rank exactly $\#X_{j,s}(j-1)$, and the alternating combination is a *selector*, not a source of cancellation. The effective dimension (distinct Frobenius eigenvalues), computed by two independent methods that agree on every row — factoring Galois-orbit products over $\mathbb{Q}$, and Berlekamp–Massey on $n \mapsto T_{j,s}(n)$

— is 172/192 at $(j,s)=(7,3)$, 972/1024 at $(9,4)$, 5004/5120 at $(11,4)$, 11086/11264 at $(12,4)$, 24274/24576 at $(13,4)$; the absolute dimension grows by a factor 2.19 to 2.33 per unit of j, i.e. $\sim 2^{j-2}(j-1)$, and the formal bound is *attained exactly* at several small cells, so no dimension count below $\#X(j-1)$ is even true. (**FINITE**; 107/107 agreements with brute force at $j=3..7$ over $\mathbb{F}_2, \mathbb{F}_4, \mathbb{F}_8$.) Katz's own words on the published state of the art: "At present, we do not know uniform bounds for these sums of Betti numbers $C(p,n,\Xi)$ as p varies."

2. Horizontal Sato–Tate at fixed $\mathbb{F}_2$ via automorphy (PROVED circular). Weyl's criterion at frequency n for this family is *literally* orthogonality: $\sum_{\chi\in X}\sum_i \alpha_\chi^n$ is the four-population integer. So "horizontal Sato–Tate at frequency n" is the target restated, not an input. Two exact facts kill the frequency-uniform and smoothed variants outright. First, $T_{j,s}(m)=+\#X_{j,s}$ for *every* odd $m<j$ (171/171 instances, $j=4..16$, all layers), so (HWO)'s inequality is *false* at low frequencies and the target is frequency-selective, living only at $m \gtrsim j$. Second, smoothing is exactly self-defeating: Fejér windows of width H reduce $|w|$ by the independent-frequency prediction and de-smoothing costs exactly H — for the admissible Weil polynomial Q of Barrier I, the $H=9$ Fejér average is $0.0786 = 0.7071/9$ on the nose.

3. An arithmetic uncertainty principle: 2-adic rigidity forcing archimedean decorrelation (PROVED refuted). The obstruction is a lemma. For $0\neq\beta\in\mathbb{Z}[\zeta_{2^s}]$ the product formula gives $\max_\sigma|\sigma\beta|\geq 2^{v(\beta)}$, and the conjugates of $S_n(\chi)$ are the $S_n(\chi^u)$, which lie in the *same* exact layer: so 2-adic divisibility of one member certifies archimedean *largeness* of another (zero violations over 285 characters). Rounding runs the same way: $T\in\mathbb{Z}$, $|T|\leq W$, $2^v \mid T$ gives $|T|\leq 2^v\lfloor W/2^v\rfloor$, and a factor 4ℓ saving would force $T=0$. 2-adic rigidity is the property of being *constant* on the layer, and a constant separates nothing. Measured, over 16,646,144 exact twisted cylinder sums at $(24,49)$: $\operatorname{corr}(v_2(A_\psi), \log_2|A_\psi|)=+0.003$, the conditional mean flat to 0.7% across $v_2=0..8$, and the v_2 histogram halving exactly.

4. Manifest positivity (PROVED one factor short). There is a genuinely new exact identity here. With $h=n-\ell$ and $L_d=\sum_{F\in W_n}\sum_{Q=P^k,\deg Q=d,\ Q|F}\Lambda(Q)\geq 0$, Type-I exactness (Lemma 3.4) gives $L_d=2^{n-\ell}$ for every $d\leq h$, hence

$$\sum_{d=h+1}^{n} L_d = \ell\, 2^{n-\ell} \quad \text{(exact, all terms } \geq 0), \qquad N_\ell(1) = 2^{n-\ell} - \sum_{d=h+1}^{n-1}\left(L_d - 2^{n-\ell}\right),$$

verified on 36 pairs (ℓ,n), $\ell=3..20$. It converts the needed lower bound into an upper bound on $\ell-1$ sparse-class prime counts, i.e. into something Brun–Titchmarsh-shaped. What it lacks, exactly: the target needs relative accuracy $1/(2(\ell-1))$ on the tail (measured 0.0263 at $(20,41)$), and Brun–Titchmarsh caps at the constant 2 (Theorem 3.8). The tail terms cancel heavily across d ($\sum|E_d|/|\sum E_d| = 7.5,\ 35.6,\ 40.0$ at $(12,25),(18,37),(20,41)$), which is precisely what term-by-term bounds destroy. Also killed here: divisor moments are prime-free ($\sum_{F\in W_n}\tau(F)=(n+1)2^{n-\ell}$ exactly, and $\sum\tau_3$ is reproduced from ball triple correlations with no primes), and $D=N_\ell(1)-2^{n-\ell}$ changes sign across the dumps, so D itself cannot be made nonnegative.

5. Clifford-hierarchy aggregate cancellation (PROVED Gaussian above order 4). Identify a character of $\mathcal{E}_j$ with a diagonal unitary on n qubits; the Witt order 2^s is exactly the level of the Clifford hierarchy, with $s\leq 2$ the stabilizer regime (quadratic phases, the Kerdock form) and $s\geq 3$ post-Clifford. The boundary is exact and sharp: the pure $\mathbb{Z}/4$ layer $(j,s)=(2,2)$ has $S=\pm 2^{(n-1)/2}(1\pm i)$ exactly, attaining Weil with ratio 1.0000, while $|S|^2$ leaves $\mathbb{Z}$ for 46,512 of 46,592 order-≥ 8 characters at $(12,25),(14,29),(16,33)$. Above the boundary every even moment is at its complex-Gaussian value: at $(22,45)$, $M_2/(\#X(j-1)2^n)=0.995..1.004$, $M_4/(2\#X((j-1)2^n)^2)=0.96..0.98$, $M_6/(6\#X(\cdot)^3)=0.91..0.94$. Consequently Hölder from the $2m$-th moment is *worse* than Cauchy–Schwarz for $m\geq 2$, and Cauchy–Schwarz itself is short of (HWO) by $4\ell/\sqrt{2(j-1)}$ — 13.9× at $(\ell,j)=(22,21)$ and 40.2× at $(200,199)$. **The needed saving is invisible to every $|\cdot|^{2m}$.** The independent-theorem version of the same statement is Dalzell–Harrow–Koh–La Placa: the aggregate of a complete degree-3 layer is provably Gaussian.

The common wall. All five reduce to one missing statement: a *phase-alignment* (delocalization) bound at frequency $n\approx 2j$ over a complete post-Clifford layer, at fixed $q=2$ and growing conductor. The required input is genuinely first-moment — the second moment is already at its random value, and the data agree with the random model throughout. Its integer analogue is conditional on GRH plus a pair-correlation hypothesis; its function-field analogues (Keating–Rudnick [26], Katz, Sawin) are all $q\to\infty$ matrix-integral limits.

Remark 5.1 (the one exact fixed-q mechanism, and where it stops)**.** There is an exact carry law in characteristic two: for Teichmüller lifts, $[a]+[b]=\sum_k V^k[c_k]$ with $c_0=a+b$ and $c_k=ab(a+b)^{2^k-2}$, whence $T_s(a+b)=T_s(a)+T_s(b)-2T_{s-1}(ab)-4T_{s-2}(ab(a+b)^2)-\cdots$ for the Galois-ring traces (verified mod 16 on 3000 random pairs in $\mathbb{F}_{2^{13}}$). Using it as an inner sum, the derivative $D_s(c)=\sum_\alpha \zeta_{2^s}^{T_s(\alpha)-T_s(\alpha+c)}$ factors through the $-2T_{s-1}(ab)$ term into an order-$(s-1)$ Galois-ring Gauss sum. Measured: $|D(c)|=0$ *exactly* for all c at $s=2$ (Kerdock), and $\approx 0.75\cdot 2^{n/2}$ at $s=3$, 0.8–0.95 $\cdot\, 2^{n/2}$ at $s=4$ — full Weil magnitude. So the carry formula collapses to Weil above the Kerdock level, with the boundary pinned exactly at $s-1=1$ (PROVED, note 07).

6 The geometric face, re-posed

This is the section a specialist should read. It re-poses (HWO) as a question about the compactly supported cohomology of one explicit sheaf on one explicit space, states exactly how much needs to be proved, records what is unconditionally true at $p=2$, and reports the computations that make the route look alive.

6.1 The space, the quotient, and the bookkeeping

Proposition 6.1 (explicit basis; PROVED, note 14 Prop. A)**.** *Let $q=2^r$, z a generator of $\mathbb{F}_q^\times$, and for k odd, $k\le j$, $0\le l<r$ put $h_{k,l}=1+z^lx^k\in\mathcal{E}_j$. Then $\mathcal{E}_j=\prod_{k\text{ odd}\le j}\prod_{l<r}\langle h_{k,l}\rangle$ with $\operatorname{ord}(h_{k,l})=2^{e_k}$, $e_k=\lfloor\log_2(j/k)\rfloor+1$ and $\sum_k e_k=j$; χ has exact conductor j iff some exponent of χ in the block $k_0=$ (odd part of j) is odd; and $\#\mathrm{Prim}_j(\mathbb{F}_q)=q^{j-1}(q-1)$.*

In characteristic two $(1+ax^k)^2=1+a^2x^{2k}$ exactly, which is what makes the orders clean and the whole computation below exact. For $j\le 7$ we have $e_1=3$, so every character value lies in μ_8 and every quantity is an element of $\mathbb{Z}[\zeta_8]=\mathbb{Z}[T]/(T^4+1)$ in integer arithmetic. Proposition 6.1 is an elementary re-derivation of Katz's $\mathrm{Prim}_j\cong\mathbb{G}_m\times\mathbb{A}^{j-1}$: the k_0-block is $W_{e_{k_0}}(\mathbb{F}_q)$, "some exponent odd" is "the initial Witt component is a unit", and $W_e^\times=\mathbb{G}_m\times\mathbb{A}^{e-1}$.

Proposition 6.2 (when the $\mathbb{G}_m$-action is free; PROVED, note 14 Prop. B)**.** *For $t\in\mathbb{F}_q^\times$ of order o, $\#\{g\in\mathcal{E}_j:\sigma_tg=g\}=q^{\lfloor j/o\rfloor}$, and the fixed characters have exact conductor j only when $o\mid j$. Hence*

$$\textit{the } \mathbb{G}_m\textit{-action on } \mathrm{Prim}_j(\mathbb{F}_q) \textit{ is free} \iff \gcd(j,q-1)=1 \iff \gcd(j_{\mathrm{odd}},2^r-1)=1,$$

and over $\overline{\mathbb{F}_2}$ it is free iff j is a power of two. When it is not, the non-free locus Z has $\dim Z\le\lfloor j/3\rfloor$.

This corrects note 12, which guessed the criterion as "$j\mid q-1$"; the two differ, and at $(j,r)=(6,2)$ there really are 12 stabilised exact-conductor characters. The correction is not load-bearing for the argument, because the non-free locus is harmless where it matters: excision gives $H_c^i(U,\mathcal{F})=H_c^i(\mathrm{Prim}_j,\mathcal{F})$ for $i>2\lfloor j/3\rfloor$, and $2\lfloor j/3\rfloor<j+1$ for every $j\ge1$, so every statement about degrees $\ge j+1$ may be made on the free open U.

Leray along the torsor. Let $\pi:U\to B:=U/\mathbb{G}_m$, a $\mathbb{G}_m$-torsor with B smooth of dimension $j-1$. Assume

> **(D)** $\mathcal{F}:=\Xi_n(L_{\mathrm{univ}})|_U=\pi^*\mathcal{G}$ for a virtual lisse $\mathcal{G}$ on B of virtual rank $j-1$.

(D) is PROVED at $j=2$ and is natural in general: the trace function is $\mathbb{G}_m$-invariant by Proposition 4.12, so $t^*\mathcal{F}$ and $\mathcal{F}$ have isomorphic semisimplifications by Chebotarev; upgrading this to an equivariant structure is what descent needs, and it would follow from $\mathbb{G}_m$-equivariance of Katz's $\mathcal{L}_{\mathrm{univ}}$. *We do not prove it.* Under (D), with $R\pi_!\overline{\mathbb{Q}}_\ell=\overline{\mathbb{Q}}_\ell[-1]\oplus\overline{\mathbb{Q}}_\ell(-1)[-2]$ and the projection formula,

$$H_c^i(U,\mathcal{F})=H_c^{i-1}(B,\mathcal{G})\oplus H_c^{i-2}(B,\mathcal{G})(-1),\quad A_r=(q-1)\sum_{b\in B(\mathbb{F}_q)}\mathrm{Tr}(\mathrm{Frob}_b\mid\mathcal{G}),\quad C=2C',\quad i_{\max}=i'_{\max}+2. \tag{2}$$

The direct sum has no cancellation, so the last equality is an equality. Artin vanishing upstairs gives $i'_{\max}\ge j-1$, hence $i_{\max}\ge j+1$: *the $\mathbb{G}_m$ factor costs exactly two degrees*, which is Proposition 4.12 re-derived.

Lemma 6.3 (connectivity is the whole content; PROVED under (D), note 14 Lemma C)**.** *With $k_{\min}=\min\{k:H^k(B\otimes\overline{\mathbb{F}_2},\mathcal{G}^\vee)\neq0\}$, Poincaré duality on the smooth $(j-1)$-fold B combined with (2) gives $2j-i_{\max}=k_{\min}$.*

6.2 The budget in the range (HWO) actually uses

Note 12 read Proposition 4.11 as demanding concentration in degree $j+1$. In the range (HWO) uses — $a \le j \le \ell$ with $a = \ell - \lceil \log_2 \ell \rceil - 1$, so $\ell/(j-1) \to 1$ — that target is set far too high.

Proposition 6.4 (the budget, restated; **PROVED**, note 14 Prop. 1′)**.** *In the range $a \le j \le \ell$, (HWO-agg) follows from Deligne's bound as soon as*

$$\delta(n,j) \;\le\; 2j - 6.15 - 2\log_2 C(2,j,\Xi_n), \qquad \delta := w_{\max} - n,$$

uniformly for $200 \le \ell \le 1024$ (the constant is 6.01–6.15 over that range, and $2\log_2 8 = 6$ in the limit). In particular a Betti sum polynomial in j, $C = j^A$, needs only $\delta \le 2j - 6.15 - 2A\log_2 j$.

So what is needed is that the top *six or seven* cohomological degrees vanish — not concentration in degree $j+1$, which is far stronger once $j \ge 8$. Two consequences worth stating explicitly.

- **The small rows cannot satisfy the budget whatever happens on them.** At $(n,j) = (7,3)$ we have $\ell = 3$ and the requirement is $k \ge 6.9$ for $C = 1$, against $2j = 6$; even cohomology concentrated in the *lowest* possible degree would give $k = 3$. Room first appears when $j - 1 \ge 6.15 + 2\log_2 C$, i.e. $j \ge 8$ for $C = 1$ — below every j we can compute. Small resolved cells are therefore evidence about the *shape*, never counterexamples to the estimate.
- **δ, not $i_{\max}$, is what the experiment sees**, and $\delta \le i_{\max}$ always, so a weight statement suffices.

6.3 What is unconditionally true at $p = 2$

Theorem 6.5 (monodromy transition, **PROVED**)**.** *$G_{\rm geom}(L_{\rm univ})$ on Prim_j contains $SL(j-1)$ for every prime p and every $j \ge 3$ except $(p,j) = (5,3)$ and $(2,3)$ (Katz [11], Thm. 5.1). So characteristic two is fully covered from $j = 4$ on. The exception is settled: at $(p,j) = (2,3)$ the normalised eigenvalues are roots of unity of order dividing 24, so $G_{\rm geom}$ is finite (Gorodetsky [13], Lemma 3.5, confirming a suspicion Katz states as Remark 5.2).* ***The transition is at $j_0 = 4$ and it is a theorem.***

This corrects a claim in our own diary, which attributed the hypothesis $p > 2n-1$ to Katz's monodromy theorem. That hypothesis belongs to his Theorem 8.2, the Betti/Weyl-sum bound, and exists only as a substitute for the missing uniform Betti bound. Katz handles $p = 2$ in Theorem 5.1 without reducing Witt characters to ordinary Artin–Schreier sheaves.

Lemma 6.6 ($H_c^{2j} = 0$; **PROVED** here, note 14 Lemma D)**.** *Let $j \ge 4$ and $n \ne j-1$. Then $H_c^{2j}(\mathrm{Prim}_j \otimes \overline{\mathbb{F}_2}, \Xi_n(L_{\rm univ})) = 0$ and $H_c^{2(j-1)}(B \otimes \overline{\mathbb{F}_2}, \mathcal{G}) = 0$; equivalently $k_{\min} \ge 1$. In particular the "no cancellation at all" case $i_{\max} = 2j$, which does occur at $(n,j) = (8,2)$ and $(12,2)$,* ***cannot occur past the transition****, and $i_{\max} \le 2j-1$ with equality iff $H^1(B \otimes \overline{\mathbb{F}_2}, \mathcal{G}^\vee) \ne 0$.*

Proof. $H_c^{2j}(X,\mathcal{F}) = \mathcal{F}_{\pi_1^{\rm geom}}(-j)$ for X smooth of dimension j. By Theorem 6.5, $G_{\rm geom} \supseteq SL_N$ with $N = j-1$ at $p = 2$, $j \ge 4$. The n-th Adams operation decomposes into hooks, $\psi^n = \sum_{k=0}^{N-1}(-1)^k s_{(n-k,1^k)}$, each once; and $(S^\lambda V)^{SL_N} \ne 0$ only for rectangular $\lambda = (m^N)$, while a hook with $k+1$ rows is rectangular only if $n - k = 1$ and $k + 1 = N$, i.e. $n = N = j-1$. Coinvariants for a group containing SL_N are a quotient of the SL_N-coinvariants, hence zero for every constituent. For B: $\pi_1^{\rm geom}(U) \to \pi_1^{\rm geom}(B)$ is surjective and $\mathcal{F} = \pi^*\mathcal{G}$, so the two have the same monodromy image and the same (co)invariants. □

Lemma 6.6 is the single piece of the degree statement that is unconditional at $p = 2$, and it is exactly the piece Sawin [16] and Gorodetsky–Sawin [15] use in their settings: "top cohomology vanishes" is $H_c^{2\dim}$ and nothing beyond. *The gap between $k_{\min} \ge 1$, which is proved, and $k_{\min} \ge 6.15 + 2\log_2 C$, which is needed, is the whole remaining problem.*

6.4 The computations

An exact L-function engine replaces the window scan. For χ a character of $\mathcal{E}_j$, $F \mapsto \chi(\langle F \rangle_j)$ is completely multiplicative, so $L(\chi,T) = \sum_{m<j} c_m(\chi)T^m$ with $c_m(\chi) = \sum_{v \in V_m}\chi(v)$ over $V_m = \{1 + b_1x + \cdots + b_mx^m\}$; computing c_m for *all* q^j characters at once is one mixed-radix Fourier transform over $\prod(\mathbb{Z}/2^{e_k})^r$ with 8th-root-of-unity kernels, hence exact in $\mathbb{Z}[\zeta_8]$. The cost is $\sim jq^j \log q$ and, crucially, is **independent of** n: every n comes out

j	n	δ	C	shape	j	n	δ	C	shape
2	5	3	2	top $(2j-1)$	3	9	3	1	j
2	8	4	1	top $(2j)$	3	11	5	2	$2j-1$
2	12	4	2	top $(2j)$	3	12	6	2	$2j$
3	7	5	1	$2j-1$	4	7	5	6	$\mathbf{j+1}$
3	8	6	2	$2j$	5	5	5	4	$\mathbf{j}$
5	6	6	4	$\mathbf{j+1}$	6	6	6	2	$\mathbf{j}$
6	8	6	2	$\mathbf{j}$	6	9	7	4	$\mathbf{j+1}$
7	7	7	3	$\mathbf{j}$	7	8	8	1	$\mathbf{j+1}$
5	12	8	6	$2j-2$	(the one exception; see text)				

Table 5: Exactly resolved cells (FINITE; note 14 §7.1, `lemire_horizontal_quotient.py`). "Resolved" means an exact closed form $A_r = (q-1)q^k P_{r \bmod m}(q)$ with integer P, verified on at least one r beyond those used to solve; $\delta = w_{\max} - n$ is then exact and C is a value, not a bound. **Every exactly resolved row with $j \le 3$ and $n \ge 2j+1$ sits at $2j-1$ or $2j$; every one with $j \ge 4$ sits at j or $j+1$**, in nine of the ten such rows. Note 12's own unresolved row $(7,4)$ is in fact $\delta = j+1$ with $C = 6$.

of one run, against $q^{n-j+1} = q^{j+2}$ for the window scan on the critical line. Cells reached: $j = 2$ ($r \le 10$), 3 (≤ 8), 4 (≤ 8), 5 (≤ 7), 6 (≤ 5), 7 (≤ 4).

Nine controls gate the engine, including exact Weil per character ($|c_{j-1}(\chi)|^2 = q^{j-1}$ and the functional equation, in $\mathbb{Z}[\zeta_8]$), integrality of A_r, the divisibility $(q-1) \mid A_r$, agreement with the independent window-scan engine on 136 overlapping (n,j,r) cells, and byte-identical output under five different blockings of the transform. Six mutation controls each die through a different named check. One control earned its keep immediately in the earlier engine: a merely irreducible (not primitive) field modulus at $r = 8$ silently zeroed most of a log table, which assert$(x^{q-1} = 1)$ does not detect but the $\mathbb{G}_m$-divisibility check does; the bad dump is kept and the checker still fails on it.

The critical line. Writing $\delta_r = 2\log_2 |A_r(n,j)|/r - n$, which converges to δ from above with bias $+2\log_2(C')/r$:

j	n	δ_r, $r = 1, \ldots, r_{\max}$	$j{+}1$	$2j{-}1$	$2j$
3	7	3.00 4.59 4.87 4.95 4.98 4.99 5.00 5.00	4	5	6
4	10	4.00 6.67 6.16 6.08 5.22 5.34 5.37 5.14	5	7	8
5	11	3.64 7.66 7.87 7.31 7.22 7.40 7.33	6	9	10
5	12	6.64 7.04 7.97 7.90 8.00 7.99 8.00	6	9	10
6	13	9.00 7.78 8.03 8.16 8.19	7	11	12
7	16	9.40 10.70 10.62 10.29	8	13	14

Taking the larger of the two critical n at the largest available r and regressing on $j = 2..7$ gives

$$\delta(2j+1 \text{ or } 2j+2,\ j) \;=\; (1.298 \pm 0.193)\, j + \text{const.}$$

The $j+1$ law has slope 1 (within 1.5σ); the $2j-1$ law has slope 2, which is $\mathbf{3.6}\sigma$ away — and the fitted slope is biased *upward*, since the available r falls as j rises. None of the four $j \ge 4$ critical rows comes near $2j-1$: at $j = 6$, 11 against a measured 8.19; at $j = 7$, 13 against 10.29. (MEASURED.)

What is *not* claimed. The critical rows at $j \ge 4$ are not resolved to an exact closed form; the affordable r is below what the fit needs once $C \ge 4$. The evidence that they are in the $j+1$ shape is (i) the exactly resolved rows of Table 5 at the same j, (ii) the growth estimates, and (iii) Lemma 6.6, which excludes $2j$ outright. And one resolved cell goes the other way: $(n,j) = (12,5)$, which closed only after the final $r = 7$ run (2^{35} group elements, 17,401 s) with one spare data point, sits at $\delta = 8 = 2j-2$ — above the $\mathbb{G}_m$-forced optimum, below the $2j-1$ law, and in the residue class $n \equiv 0 \bmod 4$ that carried the $2j$ classes at $j = 2$. It does not move the regression, but it sharpens the honest statement: at $j = 5$ the critical line is not yet in the $j + O(1)$ shape, and the case for "alive" rests on the slope across j plus Lemma 6.6, not on any single cell.

The $q = 2$ data, and what they do and do not show. The q-aspect cannot be pushed to $j \approx 200$; the $\mathbb{F}_2$ layer dumps can. Summing the exact-order layers to conductor sums $A_j = \sum_s T_{j,s}$ and putting $\delta_1(n,j) = 2\log_2(|A_j|/2^{n/2})$, a regression over the 56 points with $14 \le j \le 24$ from five (ℓ, n) pairs gives

$$\delta_1(j) \;=\; (1.003 \pm 0.144)\, j + 2.30, \qquad \text{residual sd } 2.64,$$

with slope 2 lying 6.9σ away. Three honest qualifications, in order of importance. (i) It **refutes** hypothesis (H) of note 12 — that the semisimplification is geometrically $\pi^*(M^{\otimes n})$ for a rank-one M on $\mathbb{A}^1$, which would force $i_{\max} = 2j-1$ and $C = 2$ for all j. (H) is an exact identity $|A_r(n,j)| = q^{j-2}|A_r(n,2)|$, so it is refutable at $q = 2$, and it over-predicts by 30 to 100× at $j = 18..24$. With (H) goes the only concrete mechanism note 12 could propose for a general obstruction. (ii) A $q = 2$ measurement bounds $i_{\max}$ from *below*, never above: a large $i_{\max}$ whose class happens to cancel at $r = 1$ would be invisible. (iii) Slope 1 is also the square-root-cancellation value, so this is substantially the statement "at $q = 2$ the conductor sums are of square-root size", which the earlier notes already report. The *upper* half of the argument can come only from the q-aspect, where the uncancellable factor $q-1$ makes δ a genuine measurement.

6.5 The two open statements

(T1) The degree statement (OPEN). For $n \in \{2\ell+1, 2\ell+2\}$ and $a \le j \le \ell$,

$$H^k(B \otimes \overline{\mathbb{F}_2}, \mathcal{G}^\vee) = 0 \quad \text{for all } k \le 5.15 + 2\log_2 C, \qquad \text{i.e.} \quad k_{\min} \ge 6.15 + 2\log_2 C.$$

Equivalently, by Lemma 6.3, $i_{\max} \le 2j - 6.15 - 2\log_2 C$. Full middle concentration on B ($i_{\max} = j+1$) is $k_{\min} \ge j-1$, which is far more than is needed once $j \ge 8$.

(T1w) The weight statement, which is what the estimate really needs and what the experiment measures (OPEN): the top Frobenius weight in $H_c^*(\mathrm{Prim}_j \otimes \overline{\mathbb{F}_2}, \Xi_n L_{\mathrm{univ}})$ is at most $n + 2j - 6.15 - 2\log_2 C$. (T1) implies (T1w).

(T2) The Betti statement (OPEN): $C = 2C' \le (j-1)2^{j-1-i_{\max}/2}/(4\ell)$; with $i_{\max} = j+1$ this is $C' \le (j-1)2^{(j-1)/2}/(16\ell)$.

statement	status
$\mathrm{Prim}_j \cong \mathbb{G}_m \times \mathbb{A}^{j-1}$, $\#\mathrm{Prim}_j(\mathbb{F}_q) = q^{j-1}(q-1)$	PROVED
$\mathbb{G}_m$ free $\iff$ $\gcd(j, q-1) = 1$; non-free locus dim $\le j/3$	PROVED (corrects note 12)
$i_{\max} \ge j+1$ ($\mathbb{G}_m$ costs two degrees)	PROVED under (D)
$2j - i_{\max} = k_{\min}$	PROVED under (D)
$G_{\mathrm{geom}} \supseteq SL_{j-1}$ at $p = 2$ for $j \ge 4$	**theorem** (Katz)
G_{geom} finite at $(2,3)$, eigenvalues in μ_{24}	**theorem** (Gorodetsky)
$H_c^{2j} = 0$, i.e. $k_{\min} \ge 1$, for $j \ge 4$, $n \ne j-1$	PROVED (Lemma 6.6)
$k_{\min} \ge 6.15 + 2\log_2 C$ — (T1)	OPEN
$\delta \le 2j - 6.15 - 2\log_2 C$ — (T1w)	OPEN
$C \le (j-1)2^{(j-1)/2}/(8\ell)$ — (T2)	OPEN, best in print short by an exponential
hypothesis (H) of note 12	**refuted**

Verdict. Alive, not proved. The top-degree classes of the small resolved rows are a finite-monodromy artefact of $j \le 3$ and do not persist past $j_0 = 4$; the case $i_{\max} = 2j$ is impossible there by Lemma 6.6; in every cell we can resolve the measured top weight above n is $j + O(1)$, never $2j - O(1)$; and the budget asks only that the top seven degrees vanish. Confidence: moderate-to-high on "not dead" — the route is closed by no evidence we have, and note 12's proposed obstruction is refuted; low-to-moderate on "provable", because what blocks it is now two named open statements, both open in the literature in forms strictly weaker than what is needed here.

7 What might still work

Ranked by our estimate of expected value, each with the precise obstacle. None of these is a plan for a proof; they are the places where the obstacle is identified and bounded rather than diffuse.

(i) The decisive computation

Extend the L-function engine from $\mathbb{Z}[\zeta_8]$ to $\mathbb{Z}[\zeta_{16}]$ — i.e. from $j \le 7$ to $j \le 15$ — and measure $\delta(2j+1, j)$ and $\delta(2j+2, j)$ for $j = 8, 9, 10$ at the largest affordable r. The cells $(j, r) = (8,4), (9,3), (10,3)$ all have $q^j \le 2^{32}$, the size of the $(4,8)$ cell already run, so this is a bounded engineering task rather than a research one. It doubles the j-range over which δ is a genuine measurement and directly separates $\delta = j + O(1)$ from $\delta = 2j - O(1)$ — the alive law from the dead one.

This produces a number, not a theorem, and it must be said plainly that no amount of it proves anything: the range needed is $j \geq a \approx \ell \geq 200$. Its value is decision-theoretic. If δ turns upward toward $2j$ past $j = 7$, the horizontal route is dead by Proposition 4.11 and the effort belongs elsewhere; if it stays at $j + O(1)$ through $j = 10$, past the monodromy transition and past the one anomalous cell $(12, 5)$, then (T1w) is the right thing to work on. Everything else in §6 is either settled or needs a theorem.

(ii) (T1) as the characteristic-two case of Sawin's Hypothesis H

Sawin [17] defines, on the same space Prim_n (his n is our j),

> **Hypothesis** $H(n, r, \tilde{r}, w)$: for all such $\mathcal{F}$, $H^i_c(\mathrm{Prim}_{n,\overline{\mathbb{F}_q}}, \mathcal{F}) = 0$ for all $i > n + w$.

Since $\dim \mathrm{Prim}_n = n$, this is $i_{\max} \leq n + w$, i.e. $k \geq n - w$, and **Proposition 6.4 is exactly Hypothesis H with** $w = j - 6.15 - 2\log_2 C$ — the *weakest* nontrivial case, w just below the trivial $w = j$. (Full middle concentration is $w = 1$.) His own remark: "the larger w is, the weaker an assumption Hypothesis 1.3 is." His main moment theorems are conditional on it, and he writes that there is "some hope for a geometric proof of this cohomological hypothesis".

The unconditional partial result is his Lemma 5.3, giving $H(n, r, 1, w)$ with $w = n + 1 - \big((p - 2r)/p^r\big)n$. Its proof opens: "We may assume $p > 2r$ as if $p \leq 2r$ then the claim follows immediately from the fact that $H^i(\mathrm{Prim}_{n,\mathbb{F}_q}, \mathcal{F}) = 0$ for $i > 2n$ because $\dim \mathrm{Prim}_n = n$."

> **At $p = 2$, $p \leq 2r$ for every $r \geq 1$, so Lemma 5.3 is vacuous — while at $p \geq 3$, $r = 1$ it already gives $w \approx 2n/p + 1 \leq n - 7$ for $n \geq 24$, which is more than enough.**

That is a striking coincidence of obstacles: *for every odd p the horizontal route's degree requirement is already a theorem in the relevant range*, and Kaser–Lemire is a $p = 2$ question. The same lemma is the one that blocks Sawin's Waring-problem paper [20] at $p = 2$; two independent papers die at the same place. The underlying loss is a $\lfloor n/p \rfloor$ in the singular-locus estimate of [18] Prop. 2.5, which is weakest exactly at $p = 2$. This is a bounded, identified technical gap in one lemma, not a missing theory — which is the reason it heads this list among the theorem-shaped items.

Two caveats. Sawin's $\mathcal{F}$ are summands of $\det^{-1}(L_{\mathrm{univ}}) \otimes \bigotimes \Lambda^{d_i}(L_{\mathrm{univ}})$ with $d_i \leq n - 1$; our $\Xi_n(L_{\mathrm{univ}})$ is an alternating sum of hooks with $n \approx 2j$, which lie in his class only with $r \approx 2j$, and then $p > 2r$ is out of reach anyway. So what transfers is the *shape* of the hypothesis on the same space, not a citation. And a weight statement (T1w) would do just as well as the degree statement, without any vanishing — a purity or perversity argument bounding the top weight by $n + 2j - 7$ suffices.

(iii) (T2): uniform-in-p Betti bounds

The best bound in print that is uniform in p is Sawin [17] Lemma 2.11: for $0 \leq d_1, \ldots, d_{r+\tilde{r}} \leq n - 1$,

$$\sum_j \dim H^j_c\Big(\mathrm{Prim}_{n,\mathbb{F}_q}, \bigotimes_{i \leq r} \Lambda^{d_i}(L) \otimes \bigotimes_{i > r} \Lambda^{d_i}(L^\vee)\Big) \;\leq\; 4\big(2 + \max(r, \tilde{r})\big)^{n + \sum d_i}.$$

Nothing in it depends on p. Writing a hook $S^{(n-k,1^k)}(L_{\mathrm{univ}})$ as a summand of $\bigotimes_{i=1}^{n-k} \Lambda^{d_i}(L_{\mathrm{univ}})$ gives $r = n - k$ and $\sum d_i = n$, hence $C = 2^{O(j \log j)}$ on the critical line — against an allowed $C \lesssim (j-1)2^{(j-1)/2}/(8\ell) \approx 2^{j/2}$ *even under full middle concentration*. So (T2) is a genuinely separate requirement and the incumbent misses by an exponential factor.

The other live route is Hu–Teyssier [24], which bounds Betti numbers of *wildly ramified* local systems by a polynomial in the highest logarithmic conductor times the rank, in any dimension and any characteristic, and *graded by cohomological degree*: $h^i(\mathbb{A}^n, L) \leq b_i(\mathrm{lc}_H(L))\,\mathrm{rk}(L)$ with $b_0 = 1$, $b_1 = x$ and an explicit recursion for b_n. That is four properties (T2) wants — uniform in p, uniform in ℓ, graded, linear in the rank — and the incumbent has neither of the last two. Evaluating their recursion:

k	0	1	2	3	4	5	6	7	8
$b_k(1)$	1	1	17	184	2459	39205	733496	15815563	386910089
$b_k(1)2^{-k/2}$	1	0.71	8.5	65	615	6931	$9.2{\cdot}10^4$	$1.4{\cdot}10^6$	$2.4{\cdot}10^7$

gives $\log_2 b_n(1)/n = 3.8, 4.7, 5.2, 5.6$ at $n = 10, 20, 30, 40$ against $\log_2 n = 3.3, 4.3, 4.9, 5.3$: $b_n(1) = 2^{\Theta(n \log n)}$, the same order as the incumbent. And the graded form does not rescue it, because the Deligne budget weights degree $2j - k$ by $2^{-k/2}$ and the partial sums $\sum_{k \le K} b_k(1) 2^{-k/2} = 1,\ 1.71,\ 10.2,\ 75,\ 690,\ 7621, \ldots$ *diverge* against a budget of $1/4$. So the target is now sharper than "a polynomial Betti bound": it is to **reduce the coefficients** $17, 184, 2459, \ldots$ **of** b_k **for** $k \le 7$, whose *degrees* are already optimal. (Note 15 §2.5; this replaces note 14's assessment of the same paper as "the most promising unexplored input for (T2)" — it is now explored, and it does not close it.)

Downstream note: under (T1), $C' = |\chi_c(B, \mathcal{G})|$ is an Euler characteristic, and T. Saito's index formula computes exactly that. But it computes an alternating sum, so it becomes a Betti bound only *after* concentration is known: it is downstream of (T1), not an alternative to it.

(iv) The Type II bilinear face

By Proposition 4.10, a Type II estimate for the window is a bound on $\sum_{\chi \ne 1} A_M(\chi) B_L(\chi)$ with saving n^{-A}, uniform in the coefficients, for $M, L \ge \delta n$. Fouvry–Kowalski–Michel–Sawin [25] prove Polya–Vinogradov-range bilinear bounds $\sum_m \sum_n \alpha_m \beta_n K(m^b n^c) \ll \|\alpha\|_2 \|\beta\|_2 (MN)^{1/2 - \eta}$ for the trace function K of a middle-extension sheaf whose geometric monodromy acts irreducibly with either simple identity component or finite quasisimple image — a *qualitative* monodromy hypothesis rather than "this is a Kloosterman sheaf".

The monodromy hypothesis is already a theorem here: by Theorem 6.5, $G_{\mathrm{geom}}(L_{\mathrm{univ}}) \supseteq SL_{j-1}$ at $p = 2$ for every $j \ge 4$, which is irreducible with simple identity component. The blockers are elsewhere: (a) their modulus is **prime**, ours is $x^{\ell+1}$; (b) their kernel is a trace function on $\mathbb{A}^1$ evaluated at $m^b n^c$, ours is a character of a ray class group; (c) their ranges are $M, N \le q/2$ with q the modulus size, and in the transplant $M + L = n$ with modulus $2^{\ell+1} \approx 2^{n/2}$, so $2^M \approx q/2$ sits exactly at the boundary — not disqualifying, but not slack either. Of these, (a) is the one that has to be attacked.

(v) A characteristic-two Möbius estimate

Every route through Bagshaw (Theorem 3.10) and through Sawin–Shusterman's level of distribution passes through their Möbius-in-progressions bound, whose proof *is* quadratic reciprocity: it needs $\mathbb{F}_q^\times$ to have a unique quadratic character, and it does not in characteristic two. So the question is blunt: **is there another route to Möbius cancellation in progressions over $\mathbb{F}_{2^r}[T]$?** We know of none, and we record two adjacent negative data points. Bagshaw's q-free bilinear Kloosterman bounds control additive characters of the inverse; the classical bridge to multiplicative characters mod π^n is Postnikov's formula $\chi(1 + \pi u) = \psi(a \log(1 + \pi u))$, and *the p-adic logarithm does not exist in characteristic two* — a clean statement of why the Kloosterman technology does not reach our characters at $p = 2$. And Klurman–Mangerel–Teräväinen [32] prove that at q even the class of characters with bounded short-interval discrepancy is *strictly larger* than at q odd: any fixed-q statement at $p = 2$ must survive a documented low-characteristic degeneracy that odd p does not have.

(vi) Things we judge dead, so nobody repeats them

- **Any moduli-only argument** (mass, positivity, Fourier moduli, low moments): Barrier I, with an explicit counterexample population.
- **Any group action or symmetry**, including the Gorodetsky–Kovaleva gcd identity in its general form: Barrier II, orbit ≤ 2, and $\sim 2j \ln j$ single-position characters against 2^j.
- **Any construction in the known irreducibility-preserving toolbox**: Barrier III, silent at prime n.
- **Any lower-bound sieve at the window's own level**: Barrier IV, with an exact rational prime-free population — and any such proof would prove Legendre for $\mathbb{F}_2[t]$.
- **Any Betti-number bound alone**: Barrier V.
- **The sparsity+Plancherel disproof of (CYL)**: Barrier VI, capped at $\sqrt{8\ell}$ against a 3×10^{25} shortfall at $\ell = 200$.
- **Second moments, higher even moments, Cauchy–Schwarz, Hölder**: every even moment is at its random value, and Hölder runs the wrong way (§5, shape 5). The estimate is genuinely first-moment.
- **The 2-adic tower** as a source of archimedean cancellation: shape 3, with the product-formula obstruction.
- **Smoothing or averaging over** n: exactly self-defeating (shape 2); and the family has $\sim \ell 2^\ell$ Frobenius angles while the useful range of n has $\sim \ell$ members, so any averaging the problem offers is over a set exponentially smaller than the family.

- **Swan/parity as a lower bound**: Swan's theorem gives the parity of the factor count; Swan-odd yet reducible in-window trinomials are abundant (first instances $(n, k, \#\text{factors}) = (10, 5, 3), (12, 1, 3), (13, 2, 3), (14, 1, 3)$), and separating $r = 1$ from odd $r \geq 3$ needs a smallest-factor bound that does not hold.
- **Hypothesis (H)** of note 12 (a rank-one geometric model for the semisimplification): refuted by the $q = 2$ layer sums, which it over-predicts by 30 to 100×.

7.1 The three questions, for a specialist

Question 7.1 (Q1′)**.** Fix $p = 2$ and let Ξ pick out the exact-order-2^s layer. Put $C(2, j, \Xi) = \sum_i h^i_c(\mathrm{Prim}_j \otimes \overline{\mathbb{F}_2}, \Xi(L_{\mathrm{univ}}))$ and $i_{\max}(2, j, \Xi) = \max\{i : h^i_c \neq 0\}$. *Is the pair admissible*, i.e. is $C(2, j, \Xi) \leq (j-1)2^{j-1-i_{\max}/2}/(4\ell)$? Equivalently, with $k = 2j - i_{\max}$: do the top k degrees vanish with $k \geq 2\log_2\big(8\ell C/(j-1)\big)$ — a *logarithmic* number of vanishing top degrees once C is polynomial? By Proposition 4.12 the extreme case $i_{\max} = j$ is unavailable; by Proposition 6.4 the honest target in the (HWO) range is $k \geq 6.15 + 2\log_2 C$, i.e. (T1)/(T1w) together with (T2). *The earlier form of this question — "is C polynomial in j?" — was the wrong question*, since a Betti bound alone can never suffice (Proposition 4.11).

Question 7.2 (Q2)**.** Does the joint monodromy of the pair $\big(L_{\mathrm{univ}},\ L_{\mathrm{univ}} \circ [x \mapsto x^{2^s}]\big)$ over Prim_j at $p = 2$ force first-moment cancellation of the layer sum — a *horizontal* (conductor-aspect, fixed field) equidistribution, rather than the vertical ($q \to \infty$) statements of Katz and Sawin?

Question 7.3 (Q3)**.** Is a fixed-q pair-correlation or variance theorem for a fixed-conductor Hayes family provable, even weakly — a power saving $2^{-\delta j}$ over the diagonal in the cylinder covariance? Its integer analogue is conditional on GRH plus a pair-correlation hypothesis; its known function-field analogues are all $q \to \infty$. Note the documented low-characteristic degeneracy at q even (§(v) above), which such a statement must survive.

An affirmative answer to any one of these closes Kaser–Lemire over $\mathbb{F}_2$ for all n. A proof must be uniform in ℓ, genuinely characteristic-two, applicable to a prime-power modulus, and first-moment — not a variance bound, which the data show is already at the random value and 40× short.

8 Methodology

This section is about how the work was done, not about the mathematics. A reader who wants only the mathematics can stop here.

The work recorded above was produced by an automated multi-agent process: a coordinator and a series of independent agents, each given one angle, required to read primary sources rather than abstracts, to test every idea against an exact computation, and to return either a statement or a precise obstruction. Every quantitative claim carries a checker — a script that recomputes it and exits nonzero on failure — and most checkers carry mutation controls, i.e. deliberate faults each of which must kill exactly one named check.

That discipline is not a stylistic preference. **It is there because our own claims were wrong repeatedly, in ways that were load-bearing and that review caught only because there was something to check against.** Five instances, all real:

1. **Two false lemmas in the construction note.** The first version of note 09 asserted that monomial composition $f(x^t)$ stays in-window only for degree-2 seeds, and gave a tail formula for non-monomial substitutions that was wrong as stated. Both were used to conclude that the construction route was exhausted at $n = 2 \cdot 3^k$. In fact monomial substitution is in-window for *every* seed (Theorem 3.3), and the correct criterion for a non-monomial σ is that the seed degree be a power of two. The note contradicted itself in the same paragraph, quoting $(x^4 + x + 1) \circ (x^2 + x) = x^8 + x^4 + x^2 + x + 1$, which is in-window. Worse: the lane's own Rust CAS had the correct criteria all along, in a function named `composition_shape_criterion` with a `least_binary_place` field. *The library was right and only the prose was wrong*, which is the failure mode that documentation-only review cannot catch.
2. **A priority claim that was fifteen years stale.** The same note called $n = 2 \cdot 3^k$ "the first proven infinite family". Emil Jeřábek gave it in the original MathOverflow thread in November 2011, and it is an exercise in Lidl–Niederreiter. The claim was withdrawn and the credit corrected in the published roadmap.
3. **A literature ceiling wrong by a power.** Note 09 v1 said the provable prescribed-coefficient ceiling at $q = 2$ is $\sqrt{n}$. That is Pollack's *arbitrary-position* bound; the ceiling for the *top* positions — which is what the window is about — is $n/2 - \log_2 n$ (Hayes/Weil, sharp form Hsu = Cohen). The error made the gap look like a power when it is a logarithm, and it pointed effort away from the analytic route rather than toward it.

4. **An open question that was provably the wrong question, published in the document specialists were pointed at.** Note 10 asked for a uniform bound on the Betti sum $C(2, j, \Xi)$ and asserted that a polynomial-in-j bound would give (HWO). Proposition 4.11 shows the implication is false: if the cohomology reaches degree $2j-2$ or above, even $C = 1$ fails. The binding constraint is the cohomological *degree*. The question was rewritten as (Q1′).
5. **Two agent-produced findings withdrawn on review.** A 2026 literature sweep returned two "levers". The first, the Gorodetsky–Kovaleva $\gcd(k, q^n-1)$ identity, was reported as landing within an absolute constant factor 4 of (HWO) instead of the usual factor ℓ. It is a real theorem and the arithmetic was right *per character*; per layer it is vacuous, because the characters it applies to are the single-position ones, of which there are $\sim 2j \ln j$ against 2^j — a 2^{-1011} fraction at $j = 1024$ — and it is mechanically the same power-map action that Barrier II already closed. The second reported that Bagshaw's level of distribution gives Kaser–Lemire over $\mathbb{F}_q$ for all large n once $q > 7101p^2$, and hence "at $p = 2$, every $q \geq 2^{15}$", reframing the residual problem as small q. Reading the source showed the paper fixes q an *odd* prime power in its standing notation and that the hypothesis is mechanism, not decoration — the Möbius input is quadratic reciprocity. There is no $p = 2$ case. Independently, the threshold is a condition on p^{l-2}, so no prime field and no $q = p^2$ qualifies at any size and the admissible set has density zero. Both halves of the claim were corrected (§3).

Two smaller corrections in the same vein are recorded above because they change numbers a reader might reuse: the reversal index is $\lceil n/2 \rceil$, not $\lfloor n/2 \rfloor + 1$, and at even n the difference can turn a nonempty progression into an empty one; and the $\mathbb{G}_m$-freeness criterion is $\gcd(j, q-1) = 1$, not $j \mid q-1$.

The general lesson is not about automation. It is that **a claim which carries a checker that can fail is a different kind of object from a claim that does not**, and that adversarial review of one's own findings — reading the source rather than the abstract, and asking of every negative result what the command would have printed if it were broken — is what kept this record honest. The two withdrawn findings above were both produced by a competent process, were both plausible, and were both wrong; what caught them was that somebody was required to check them against the primary text and against an exact computation, and had the data to do it. Where a claim here is finite evidence rather than a theorem, it says so; where it is a measurement, it says that too. We would rather this document be dull and correct.

Reproducibility. The research notes (00–17) are in `docs/research/10-cas/lemire-signed-trace/` and the checkers and exact data in `scripts/lemire-signed-trace/` of the repository [35]. The load-bearing scripts, by section: `lemire_almostall.py` (§3.2), `lemire_composition_family.py` (§3.3), `lemire_sieve_face.py` (§3.4, §4.4), `lemire_largeq.py` (§3.5), `lemire_horizontal_weights.py` and `lemire_horizontal_quotient.py` (§4.5, §6), `lemire_cylinder_plancherel.py` (§4.6), `lemire_layers.py`, `lemire_twists.py`, `lemire_cylinders.py`, `lemire_witt.py` (§5). Bulk producers are three independent engines — a Rust GF(2) computer algebra system, python-flint, and a pure-Python reference implementation — which are required to agree on every overlapping cell.

References

[1] D. Lemire, "Can we always find such an irreducible polynomial of degree n where degree$(p(x)-x^n) \leq n/2$?", *MathOverflow* question 81717, 23 November 2011. `https://mathoverflow.net/questions/81717`. (Answers and comments by N. Elkies, J. Ellenberg, E. Jeřábek, J. Lahtonen, I. Rivin, J. F. Voloch, G. Zaimi cited in the text.)

[2] O. Kaser and D. Lemire, "Strongly universal string hashing is fast", *The Computer Journal* **57**(11) (2014), 1624–1638; arXiv:1202.4961, section "GF Multilinear". (The conjecture is stated there, verified for $L \leq 400$ against [3], and motivated by Barrett reduction. The question itself is [1], three years earlier.)

[3] J. Arndt, *Matters Computational: Ideas, Algorithms, Source Code*, Springer, 2011; table `lowbit-irredpoly.txt`. (Minimal in-window subdegree is ≤ 10 for every $n \leq 400$.)

[4] D. R. Hayes, "The distribution of irreducibles in GF$[q, x]$", *Trans. Amer. Math. Soc.* **117** (1965), 101–127.

[5] A. Weil, *Sur les courbes algébriques et les variétés qui s'en déduisent*, Hermann, Paris, 1948.

[6] P. Deligne, "La conjecture de Weil. II", *Publ. Math. IHÉS* **52** (1980), 137–252. (Cor. 3.3.4 is the weight bound used in Proposition 4.11.)

[7] R. Lidl and H. Niederreiter, *Finite Fields*, Encyclopedia of Mathematics and its Applications 20, 2nd ed., Cambridge University Press, 1997. (Theorem 3.35.)

[8] G. L. Mullen and D. Panario (eds.), *Handbook of Finite Fields*, CRC Press, 2013. (Theorem 3.2.5, the single-polynomial form, after Menezes et al.)

[9] H. Fredricksen and R. Wisniewski, "On trinomials $x^n + x^2 + 1$ and $x^{8l\pm3} + x^k + 1$ irreducible over GF(2)", *Information and Control* **50** (1981), 58–63.

[10] S. W. Golomb and P. F. Lee, "Irreducible polynomials which divide trinomials over GF(2)", *IEEE Trans. Inform. Theory* **53**(2) (2007), 768–774.

[11] N. M. Katz, "Witt vectors and a question of Keating and Rudnick", *Int. Math. Res. Not.* 2013, no. 16, 3613–3638. (Thm. 5.1, the monodromy theorem, covers $p = 2$ for $j \geq 4$; Thm. 8.2, the Betti/Weyl-sum bound, is the statement carrying $p > 2n - 1$. Not on arXiv.)

[12] N. M. Katz, "Sums of Betti numbers in arbitrary characteristic", *Finite Fields Appl.* **7** (2001), 29–44.

[13] O. Gorodetsky, "Irreducible polynomials over $\mathbb{F}_{2^r}$ with three prescribed coefficients", *Finite Fields Appl.* **56** (2019), 150–187; arXiv:1805.07105. (Lemma 3.5: at $(p, j) = (2, 3)$ the normalised eigenvalues have order dividing 24, so G_{geom} is finite.)

[14] O. Gorodetsky and V. Kovaleva, "Equidistribution of high traces of random matrices over finite fields and cancellation in character sums of high conductor", arXiv:2307.01344 (2023). (`lem:sym/cor:sym`: the $\gcd(k, q^n - 1)$ symmetry.)

[15] O. Gorodetsky and W. Sawin, "Correlation of arithmetic functions over $\mathbb{F}_q[T]$", *Math. Ann.* **376** (2020), 1059–1106; arXiv:1811.04834.

[16] W. Sawin, "A representation theory approach to integral moments of L-functions over function fields" (twists by Witt-vector Dirichlet characters), arXiv:1805.04330 (2018).

[17] W. Sawin, "A representation theory approach to integral moments of L-functions over function fields", *Algebra & Number Theory* **14** (2020), 867–906; arXiv:1810.01303. (Hypothesis $H(n, r, \tilde{r}, w)$; Lemma 2.11, the uniform-in-p Betti bound; Lemma 5.3, vacuous at $p = 2$.)

[18] W. Sawin, "Square-root cancellation for sums of factorization functions over short intervals in function fields", arXiv:1809.05137 (2018). (Prop. 2.5, the singular-locus estimate that degrades by $\lfloor n/p \rfloor$.)

[19] W. Sawin, "The size of wild Kloosterman sums in number fields and function fields", arXiv:2209.02170 (2022). (The sparsity+Plancherel disproof template.)

[20] W. Sawin, "The asymptotic in Waring's problem over function fields via a singular locus in the circle method", arXiv:2412.14053 (2024). (Dies at the same Lemma 5.3 at $p = 2$.)

[21] W. Sawin and M. Shusterman, "On the Chowla and twin primes conjectures over $\mathbb{F}_q[T]$", *Ann. of Math.* **196** (2022), 457–506; arXiv:1808.04001. (Odd p, $q > 685090\,p^2$; the Möbius input is quadratic reciprocity.)

[22] W. Sawin and M. Shusterman, "Short sums of trace functions over function fields and their applications", arXiv:2512.24080 (2025). (Large fixed q, squarefree moduli, slopes ≤ 1: excluded here on all three counts.)

[23] C. Bagshaw, "Bilinear Kloosterman sums in function fields and the distribution of irreducible polynomials", *Canad. J. Math.* **78** (2026), 302–327; arXiv:2401.10399. (Cor. 2.5: level of distribution $\omega < 1/2 + 1/62$ for an arbitrary individual modulus, for **odd** p with $q > p^2 e^2((16 - \omega)/(16 - 31\omega))^2$.)

[24] H. Hu and J.-B. Teyssier, "Estimates for Betti numbers and relative Hermite–Minkowski theorem for perverse sheaves", arXiv:2502.11060 (2025).

[25] É. Fouvry, E. Kowalski, P. Michel and W. Sawin, "Bilinear forms with trace functions", arXiv:2511.09459 (2025).

[26] J. P. Keating and Z. Rudnick, "The variance of the number of prime polynomials in short intervals and in residue classes", *Int. Math. Res. Not.* 2014, no. 1, 259–288; arXiv:1204.0708. (§5.1–5.2: the reversal duality.)

[27] P. Pollack, "Irreducible polynomials with several prescribed coefficients", *Finite Fields Appl.* **22** (2013), 70–78.

[28] J. Ha, "Irreducible polynomials with several prescribed coefficients", *Finite Fields Appl.* **40** (2016), 10–25; arXiv:1601.06867.

[29] C.-N. Hsu, "The distribution of irreducible polynomials in $\mathbb{F}_q[t]$", *J. Number Theory* **61** (1996), 85–96. (Thm. 2.4.)

[30] S. D. Cohen, "Explicit theorems on generator polynomials", *Finite Fields Appl.* **11** (2005), 337–357. (Thm. 2.1, equivalent to Hsu Thm. 2.4.)

[31] J. Friedlander and H. Iwaniec, *Opera de Cribro*, AMS Colloquium Publications 57, 2010. (Thm. 11.12, the Jurkat–Richert linear sieve.)

[32] O. Klurman, A. P. Mangerel and J. Teräväinen, "Beyond the Erdős discrepancy problem in function fields", arXiv:2202.10370 (2022).

[33] M. Bommarito, *Half-degree irreducibles over $\mathbb{F}_2$: what is proved and what remains open*, 2026 (`lemire-proof-roadmap.pdf` in this repository).

[34] M. Bommarito, *Almost all half-degree patterns are realized by an irreducible over $\mathbb{F}_2$*, 2026 (`lemire-almost-all.pdf` in this repository).

[35] M. Bommarito, *Axeyum: research ledger, notes 00–17 of the `lemire-signed-trace` lane, and replayable diagnostics*, 2026. `https://github.com/mjbommar/axeyum`.